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Examination of the Effects of the Nord stream 1 and 2 Pipeline Explosions on the German Gas Market

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Bachelor Thesis Economics

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Submission date: 17.05.2024

Abstract

In this paper, I will examine the impact of the Nord Stream attack on the German gas market. Thus, I pose the research question: "How has the German gas market been affected by the attack on the Nord Stream pipelines?" I provide an answer in two steps. First, I assess the attack's impact on Germany's and Western Europe's capability to import pre-war volumes of Russian gas through a maximum flow analysis of the European gas network. Second, I employ a multiple linear regression model to estimate how various factors influenced by the Nord Stream attack have affected gas prices, thereby analysing the price impact of the attack.

Regarding import capabilities, my paper concludes that the still-existing physical pipeline infrastructure could fulfil the pre-war gas demand of Western Europe and Germany. However, the politically motivated closure of the Yamal pipeline by Poland further reduced capacities, meaning that in the situation post-attack, the attack has reduced available capacities to a level insufficient to supply pre-war import volumes. My analysis found no price-increasing effect from the Nord Stream attack, due to prior market anticipation that no Russian gas would be imported through Nord Stream even if it were fully operational. This factor led to price spikes before the Nord Stream attack, and a price reduction after the attack, when the supply-demand gap was less striking than expected by the markets due to intensive gas savings from the EU and Germany.

Overall, I conclude that the Nord Stream attack eliminated the possibility for Western Europe and Germany to independently decide to return to pre-war import volumes from Russia, without considering transit nations. Since the market had already anticipated lower import volumes from Russia, and the Nord Stream attack did not drive this process, no price-increasing effect was observed.

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Introduction

In the early hours of September 26, 2022, a series of four massive explosions occurred in the Baltic Sea, destroying three out of four pipes of the Nord Stream 1 and 2 pipelines (Soldi et al., 2023, p.4). Each pipe had an annual capacity to deliver 27.5 billion cubic meters (bcm) of natural gas (Nord Stream, 2018), which means that the attack destroyed a total of 82.5 bcm of yearly direct gas transfer capacity between Germany and Russia. This amount is equivalent to two-thirds of Norway's total annual gas production, Europe's largest natural gas producer besides Russia (Energy Institute, 2023), and nearly the entire German gas import from all sources in 2021 (BP, 2023). These comparisons highlight the enormous scale of destruction by the attack. It remains unclear at the time of the writing of this paper if the pipelines will ever be repaired (Soldatkin, Astakhova & Steitz, 2023), and it is evident that such a venture would face enormous technical difficulties and costs (Shcherban & Mazur, 2023, p.600). The capacities of the pipes destroyed remain therefore potentially permanently lost. The Nord Stream incident is in terms of scope and scale therefore an unprecedented attack on energy infrastructure, which permanently reshaped Europe's gas pipeline network.

In my opinion, the most crucial aspects of the attack are its consequences for the German gas market based on several reasons. Firstly, the significance of natural gas to the German economy cannot be overstated. Natural gas makes up almost 24% of Germany's total energy demand (Bundesministerium für Wirtschaft und Klimaschutz, 2024), and is of rising importance especially in electricity production, where its huge flexibility is used to buffer the fluctuating production of a increasing share of renewable electricity sources (Halser & Paraschiv, 2023, p.3). Maintaining secure and resilient access to sufficient natural gas supply is a priority for Germany. The Nord Stream attack, which destroyed significant direct pipeline capacities to Russia, Germany's largest source of gas supply before the war, could jeopardize this (Bundesnetzagentur, 2023). Directly related to supply security is Germany's politically controversial decision to decouple from Russian gas imports in response to Russia's full-scale invasion of Ukraine, drastically reducing its import volumes from Russia (Bundesnetzagentur, 2023). The extent to which this decision was driven and forced by reduced import capabilities due to the Nord Stream attack is therefore of significant interest. Another important aspect directly connected to the huge importance of gas for the German economy is the spiking and highly fluctuating gas prices in recent years. After the onset of Russia's full-scale invasion of Ukraine, prices spiked to heights of $340 per MW/h, which is more than 10 times the average price in 2021, and the average price in 2022 was 350% above the 2021 average (Investing.com, 2024). These high gas prices posed huge challenges to Germany's economy, making it crucial to understand the drivers of the gas price and the role of the Nord Stream attack. Assessing the direct impact on the gas market is especially important to assess the danger of potential attacks on other energy infrastructure, which is of rising relevance due to geopolitical tensions (Belyi & Piebalgs, 2024; Soldi et al., 2023, p.4).

Moreover, addressing the question of the impact of the Nord Stream attack is of significant scientific relevance from two perspectives. The factors driving the European gas price have been of considerable interest in the literature, with multiple drivers identified (Brow & Yttcel, 2008; Halser & Paraschiv, 2023; Nick & Thonenes, 2014; Obadi & Korcek, 2020; Roberts, 2019). Discussing to what extent these results still hold or have to be altered in a drastically changed infrastructure environment in Europe due

to the Nord Stream attack is therefore a valuable addition to this body of literature. Additionally, for the field of political science, the question of the impact of Nord Stream is relevant, as Nord Stream 1 and 2 were broadly and controversially discussed projects in the literature (Jong, 2023; Vosoughi & Mousaei, 2022; Wood, 2023). Especially the interests of different stakeholders, and whether they would profit or lose from the project was of great interest. An economic assessment of the impact of the Nord Stream attack could provide a valuable contribution to this discussion by providing an analysis of the counterfactual of Nord Stream 1 and 2 no longer existing.

As I have outlined, the question of the impact of the Nord Stream attack is of high relevance based on numerous aspects. Despite its relevance, papers published on the Nord Stream attack have so far only discussed its ecological consequences and lessons for the protection of undersea infrastructure (Gülcan & Erginer, 2023; Jang et al., 2022; Sanderson et al., 2023; Soldi et al., 2023). My paper aims to address this research gap by asking the research question: "How has the German gas market been affected by the attack on the Nord Stream pipelines?" I will focus on two central aspects: the impact of the attack on Germanys and western Europe's capabilities to import pre-war volumes of gas from Russia and the impact of the attack on the development of the gas price. This limits the scope of my paper while still covering the major possible impacts on the gas market. I decided to focus my analysis on Germany because it had the highest gas import volumes from Russia before the war and was directly connected to Russia via Nord Stream (Eurostat, 2023). However, since Europe's gas markets are deeply integrated, the findings are to some extent also applicable to other European nations, and some aspects are discussed more broadly where a focus on Germany would be inappropriate.

I will answer this research question by laying a theoretical foundation through a literature analysis about the Nord Stream 1 and 2 projects and the current assessments of the impact of past attacks on energy infrastructure. Based on the literature analysis, I will formulate concrete hypotheses to guide my further analysis. Next, I will analyse the impact of the Nord Stream attack on Germany's and Western Europe's ability to import pre-war volumes of gas from Russia using a max flow analysis of a graph based on Europe's pipeline network. Afterwards, I will assess the impact of multiple factors on the gas price through a multiple linear regression analysis. By operationalizing the impact of the Nord Stream attack on various variables in the model, I can assess the impact of the Nord Stream attack on the gas price. In the final step, the results of the two analysis steps will be summarized and discussed in relation to the research question, the literature, and their practical relevance.

Theory

This section will provide a thorough overview of the literature that forms the foundation of this paper. Initially, I will give an overview of the institutional context in which the two Nord Stream projects lie. Gaining a clear understanding of the institutional context, particularly the political controversy around both projects, is crucial for appreciating Nord Stream's impact on European gas infrastructure and thus essential for hypothesizing the effects of its destruction. Subsequently, I will review the literature on the specific effects of the Nord Stream attack and prior pipeline incidents to gauge their potential impact on gas prices. Together, these sections constitute the comprehensive literature foundation of this paper. I will outline the literature upon which the methods build in the two methodological parts of the analysis steps, to ensure an explicit theoretical grounding of the methods.

Theory: Institutional Context

Nord Stream 1 was announced already in 2005 and consists of two pipes, each able to deliver up to 27.5 bcm per year directly from Russia to Germany through the Baltic Sea, as it can be seen in Figure 1. It was commissioned in 2011, delivering gas until August 2022, when Gazprom temporarily halted deliveries citing technical issues (NDR, 2022). During the Nord Stream attack both pipes of Nord Stream 1 were destroyed (Soldi et al., 2023, p.4). Nord Stream 1 has been highly controversial since its announcement, being the first direct pipeline from Russia to Western Europe and bypassing all existing transit countries. These countries feared weakened negotiation leverage in tariff disputes with Russia and reduced gas transit, potentially undermining their energy security (Lochner & Bother, 2007, p.2). Furthermore, there were huge discussions about the necessity for Nord Stream 1 in terms of transfer capacity increment, because Russia would already be able to increase its exports within the existing pipeline infrastructure. Thus, many Eastern European Nations viewed Nord Stream 1 as an unnecessary route designed primarily to circumvent them (Lochner & Bothe, 2007, p.15). This is further underscored by ongoing debates and uncertainty surrounding the economic viability of Nord Stream 1 (Chyong, Noel & Reiner, 2010, p.19). Western European nations were more in favour of Nord Stream 1, mostly because they anticipated from Nord Stream 1 cheaper gas prices and increased supply security (Koczy et al., 2022, p.8). Moreover, different political views and historical experiences in the West and East with Russia caused different assessments of the fact that the Nord Stream build would result in closer economic ties with Russia (Whist, 2008, p.50-53). In the end, the supporters of Nord Stream 1 were successful, but only with the compromise that until 2017 only one of the two pipes was allowed to operate (Sziklai et al., 2020, p.2). Nevertheless,

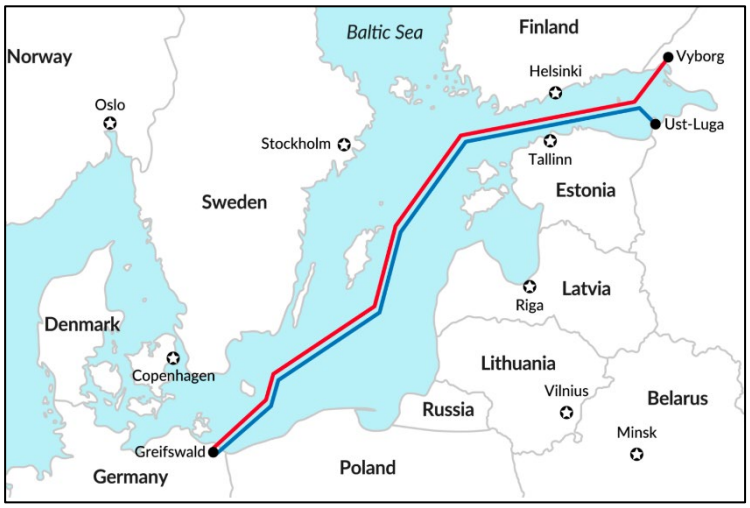


Figure 1: Route of Nord Stream 1 and 2 (Umbach, 2020)

since 2017 both pipelines of Nord Stream 1 were heavily utilized, achieving utilization rates close to 100% (IEA, 2024).

With Nord Stream 1 not even fully utilized, Vladimir Putin announced already in 2015 the building of the sister pipeline Nord Stream 2. Nord Stream 2 also consists of 2 pipes, each having a capacity of 27.5 bcm, following Nord Stream 1. Modelled after Nord Stream 1, it was constructed by a consortium of Western companies and Gazprom and became operational in 2021 (Jong, 2023, p.3). Western companies were primarily motivated by economic factors, as Gazprom assured specific transfer volumes through Nord Stream 2, ensuring a consistent cash flow. Gazprom, on the other hand, could strengthen its position on the European market with a larger direct access, and save on transfer fees compared to the land connections (Lang, Westphal & Kirsten, 2016, p.10-13). Similarly to Nord Stream 1, Nord Stream 2 raised huge controversies in the international realm among Eastern and Western European nations (Sziklai et al., 2020, p.9). Led by Germany, Western European countries viewed the project as a private economic venture, as it received no public funding. Additionally, especially Germany regarded it as an opportunity for cooperation with Russia, which was especially valuable in times of increased tensions following Russia's annexation of Crimea in 2014 (Jong, 2023, p.4). Eastern European nations, supported by the USA, were on the other side in harsh opposition to the project, fearing an even higher energy dependency of Europe from Russia. Moreover, there was concern that Nord Stream 2 would allow Russia to completely bypass Ukraine, further undermining Ukraine's strategic position during escalating tensions with Russia (Jong, 2023, p.5). This opposition led to numerous tries to halt or slow the construction of Nord Stream 2, ranging from court rulings, and permit issues to US sanctions against participating firms (Jong, 2023, p.6-8). In the end, all these attempts were unsuccessful, and Nord Stream 2 was operational in 2021 with its license for operation on the way. Only after Russia's full-scale invasion of Ukraine, did Germany stop the certification process (Jong, 2023, p.9). During the Nord Stream attack, only one out of the two pipes was destroyed, leaving one pipe ready for operations up to this date (Soldi et al., 2023, p.4).

Overall, it can be concluded that Nord Stream 1 and 2 were both highly controversial projects. The current stand of the literature is that the primary motivation for both pipelines was not to create essential infrastructure for increased transit volumes, but rather to circumvent Ukraine and other transit nations for political and economic reasons. Consequently, the bypassed transit nations were heavily opposed to both projects and economically proficient Western European nations were in favour. Ultimately, despite significant opposition, both pipelines were employed, with Nord Stream 1 seeing substantial use. Only the onset of Russia's full-scale invasion of Ukraine, and afterward the attack on Nord Stream, put an end to both projects. The extensive debates about the necessity of Nord Stream 1 and 2's transport capabilities, and its effect on gas prices, highlights the relevance of assessing, which effect the destruction of the Nord Stream pipelines had on Europe's gas infrastructure and the gas price.

Theory: Effect of pipeline attacks

This part will discuss the literature in fields relevant to this paper. Firstly, we will explore existing literature on the Nord Stream attack across various disciplines, detailing its implications for the research question. Subsequently, an overview will be given about the discussion of the impact of pipeline attacks, failures, and sabotages in the literature, furthermore the consequences for the development of hypotheses and the research method will be discussed.

Despite the Nord Stream attack's recent occurrence, there already exists scientific literature about it and its consequences. The primary focus of this literature is the ecological consequences (Sanderson et al., 2023) and the methane emissions (Jang et al., 2022) that resulted from the attack. Additionally, a few case studies have discussed the attack's implications for undersea infrastructure security. (Gülcan & Erginer, 2023; Heyburn et al., 2024; Soldi et al., 2023;) This analysis establishes an understanding of the factual details of the attack. However, to date of the writing of this paper, there has been no analysis published on the impact of the Nord Stream 1 and 2 attack on the European gas market, despite extensive prior discussions on the potential effects of these pipelines' construction on the market. This outlines the research gap that this paper aims to fill and underlines the relevance of the research question.

In the absence of literature on the Nord Stream incident's impact on gas markets, I was looking for studies examining the effects of past pipeline incidents on oil and gas markets. The case of pipeline sabotages and attacks in Nigeria stood out because it was of huge interest to the literature. Over the past two decades, a high number of incidents posed a major challenge to the oil sector of Nigeria, with approximately 374,125 barrels of oil lost between 2007 and 2015 (Onuoha, 2008, p.106; Umar, Khanan, Ahmad & Imsa'il 2019, p.246). The economic damage for the government was twofold, first of all, the government had less tax revenues because of the decreased oil sales (Umar et al., 2019, p.248), and a small but significant positive effect from the number of attacks on oil prices was found in a VAR model and a linear regression (Yeeles & Akporiaye, 2016, p.191-194). However, these findings are only on partially applicable to the Nord Stream attacks, due to differing circumstances. In Nigeria, the challenges are numerous small incidents, for instance, in 2000 alone, 909 incidents of sabotage were counted (Onuoha, 2008, p.106), whereas in the Nord Stream case, one large-scale attack happened. Additionally, even cumulatively, the scope of the attacks in Nigeria is way smaller than the effect of the Nord Stream attack, because often only small pipelines are affected, and damage can be repaired quickly. This disparity is underscored by the fact that the total energy value lost in Nigeria between 2007-and 2015 could be delivered in roughly 8 hours by capacities destroyed in the Nord Stream attack (Jong, 2023, p.3; Lochner & Bother, 2007, p.2; Umar et al., 2019, p.105).

Another significant topic in the literature concerns attacks on oil infrastructure in the Middle East and North Africa. Algeria, for example, has seen attacks from varying rebel and terrorist groups on oil and gas infrastructure, but these were only of limited scope and posed overall no significant problems for supply security and deliveries (Lacher & Kumetat, 2011, p.4476). In the Middle East, the huge attack on Saudi Aramco's oil processing plant in Abqaiq 2019 stood out, which knocked out nearly 6 million barrels of oil production per day and led to a significant price jump (Krane, 2020, p.120-121). This daily production capacity is roughly 250% of Nord Stream's daily delivery volume, which underlines the

scale of this attack. However, the capacities were quickly fully restored, and only two weeks later the oil prices were already below the pre-war values (Krane, 2020, p.120-121). So even though the scope of the effect on oil production was bigger, the effect was only short-lived and has not represented a structural change like Nord Stream did.

In conclusion, the magnitude of the Nord Stream attack is incomparable to events previously discussed in the literature. Nord Stream marks a structural break in the gas infrastructure situation of Europe, whereas, in comparison, the other discussed events can be seen as temporary shocks of a lesser magnitude. Therefore, the methodologies employed in the discussed studies are not directly applicable to analysing the Nord Stream attack but can serve as inspiration for this paper's research design. Furthermore, the observed price increases resulting from these smaller incidents provide a foundational basis for developing hypotheses concerning price trends in the Nord Stream case.

Hypothesis

In this section, I will develop hypotheses that clarify the research question, thereby laying the groundwork for the subsequent analysis chapters. The research question, which examines the impact the Nord Stream attack had on the European and especially German gas markets, can be split into two primary aspects: the impact on gas import capabilities and the impact on gas prices. The first three hypotheses will therefore focus on the impact on import capabilities, and the fourth, fifth, and sixth on the impact on gas prices.

Concerning the impact on gas import capabilities, the scale of the attack stood out. The attack effectively destroyed pipelines that could deliver 82.5 bcm of natural gas per year (Soldi et al., 2023, p. 4), which is nearly the total gas imports of Germany in 2021 (Eurostat, 2023). The 55 bcm Nord Stream 1 capacity that was destroyed in the attack had additionally high utilization rates before the attack, with transport volumes even being above the official capacities in the 3 years prior to the attack (IEA, 2024). However, as I outlined earlier in the theory part, it was always highly controversially debated whether the Nord Stream capacities were necessary to support Europe's demand for Russian gas. This assertion is supported by scholarly perspectives suggesting that the primary function of Nord Stream 1 and 2 was to circumvent transit countries on land-based routes (Lochner & Bothe, 2007, p. 15). The high utilization of Nord Stream 1 could then be explained by the attempt to bypass as much gas as possible via Nord Stream around the transit countries in Eastern Europe. Additionally, the 27.5 bcm of import capabilities of the destroyed Nord Stream 2 pipeline were never utilized and can therefore per definition have not been necessary for a pre-war import level of Europe which was achieved without them. There also still exists one Nord Stream 2 pipeline, which is technically able to deliver gas (Reuters, 2022), and could therefore substitute half of the lost previously utilized Nord Stream 1 capacities. Combining these factors leads me to the following first hypothesis:

> *Hypothesis 1: The attack on Nord Stream has not decreased the physical import capabilities of Germany and Western Europe for Russian gas to a level that makes pre-war levels of import impossible.*

The second and third hypotheses will address the fact that the political surroundings around Europe's pipeline infrastructure have changed drastically since the beginning of the full-scale invasion of Ukraine. This resulted in the closing of the Yamal pipeline by Poland in 2022 (Strzelecki, 2022), and ongoing debates about the transport permission for the trans-Ukraine pipeline which must be prolonged by Ukraine after 2024 (Gavin, 2023; Vakulenko, 2024). The closing of pipelines for political reasons has and would drastically reduce the import capabilities of Europe over alternative land routes. It seems therefore plausible to me, that in these scenarios the reduced capacities on alternative routes result in the Nord Stream attack having a significant impact on the ability of Europe to import gas from Russia on a pre-war level. Hypotheses 2 and 3 are therefore:

Hypothesis 2: The attack on the Nord Stream pipeline, compounded by the closure of the Yamal pipeline, has effectively reduced Germany and Western Europe's capacity to import Russian gas below a pre-war level.

Hypothesis 3: The attack on the Nord Stream pipeline, compounded by the closure of the Yamal and Trans Ukraine pipeline, has effectively reduced Germany and Western Europe's capacity to import Russian gas below a pre-war level.

Hypotheses 4, 5 and 6 then address the part of the research question concerned with the impact of the Nord Stream attack on the gas price. Assuming the validity of Hypothesis 2, which reflects Europe's situation during the Nord Stream attack, it is plausible that this capacity limitation led to a reduction in gas supply. Based on the literature outlined in the theory chapter, in which even smaller and shorter-lasting supply shortages caused by an attack increased prices, I assume that such a reduction in gas supply would lead to an increase in gas prices. Hypothesis 4 formulates this formally:

Hypothesis 4: The Nord Stream attack caused an increase in German gas prices due to a reduction in Russian gas imports, leading to a tightened supply and subsequent price increases across Europe.

Hypothesis 5 then addresses the factor, that such a gas supply gap caused by the reduction in import capabilities from Russia will lead to an increase of gas imports from other sources. This is especially important since the reduction of Nord Stream capabilities is up to this date a permanent reduction with currently no intent of fixing it for cost reasons (SZ, 2023). The supply gap therefore opened by the attack, under the assumption that hypothesis 2 is true, is in the case of Western Europe and Germany primarily filled with increased LNG imports from the United States (Gritz & Wolf, 2024, p. 4). But LNG, especially when it is transported a long route like LNG from the USA to Western Europe, is often cited as being more expensive than pipeline gas (Molnar, 2022, p. 54-56). The change in the supply structure of Western Europe's gas imports, and especially the increased LNG imports, could therefore lead to higher gas prices. Hypothesis 5 is therefore:

Hypothesis 5: The attack on Nord Stream caused a increase in gas prices due to a rise in the amount of imported LNG, which incurs higher transportation costs compared to pipeline gas.

The sixth hypothesis summarizes the predictions of the effects of the Nord Stream attack on gas prices. Hypotheses 4 and 5 would both positively impact gas prices if they were confirmed to be true. The attack could also raise general uncertainty about the security of the gas supply, thereby raising gas prices (Chitu, Minesso & Manu, 2024). Overall, the sixth hypothesis is that the Nord Stream attack has increased gas prices, resulting in the following formulation.

Hypothesis 6: The attack on Nord Stream caused overall a lasting increase in gas prices.

These six hypotheses cover the potential impacts of the Nord Stream attack on Western Europe's gas market and therefore specify the research question. I will now test the hypotheses in two separate analysis steps. In the first step, I will test hypotheses 1, 2, and 3 with a maximum flow analysis of the European gas network. Following the max flow analysis, I will conduct a multiple linear regression to assess the impact of various structural factors on Europe's gas prices, including the effects of total gas supply, Russian gas imports, and LNG imports. This will enable me to answer hypotheses 4, 5, and 6, and test alternative explanations and impact factors for the gas price development.

Gas Flow Analysis

In this chapter, I will present my analysis of the impact of the Nord Stream attack on the physical capacities of Western Europe to import gas from Russia at pre-war levels in different scenarios, thereby testing the first three hypotheses. I will test these hypotheses by conducting a maximum flow analysis based on a graph representing the European pipeline network in the different scenarios. This part will then provide an overview of the data sources and descriptive statistics, and afterwards explain the research design and finally analyse the consequences of the results for the hypotheses.

Gas Flow Analysis: Data

The primary data source for pipeline connections between balance zones and their physical capacity is the System Capacity Map from ENTSOG (2024). This dataset provides up-to-date information on all intra-balance zone gas transfer capacities within the EU and its neighbouring countries. However, it excludes some crucial connections for my analysis because they are currently not authorized to operate, as with the Yamal pipeline (Strzelecki, 2022). To maintain data consistency, missing data points from these connections were supplemented using older ENTSOG data (2021). Flow capacities in the ENTSOG data are given in GWh/d and then calculated to billion cubic meters per year (bcm), to match the units of yearly gas imports in official Eurostat data. Given that pipelines are consistently utilized throughout the year due to seasonal demand fluctuations being offset by gas storage, it is feasible to estimate annual capacities from daily data (European Commission, 2022, p. 17; IEA, 2024).

Figure 2 illustrates the major gas pipeline connections from Russia to Germany and therefore gives an overview over the network. Notably, the connections are on four major stages, from which three are relevant for my analysis. The Nord connection with Nord Stream 1 and 2 running trough the Baltic Sea directly from Russia to Germany. The Yamal pipeline as the middle-Nord connection trough Belarus and Poland, and the Brotherhood/Trans Ukraine pipeline tough Ukraine and then Slovakia, the Czech Republic and Austria who forms the south middle connections. The pipeline connections in the south through the black sea and the Balkans are not relevant for western Europe, because no direct connection exists to central Europe. This three-pipeline system form the backbone of gas transfers from Russia to Germany. It should be noted that the actual network and dataset complexity exceeds the simplified depiction in this overview, typically involving multiple pipelines in each pipeline system (ENTSOG, 2024).



Figure 2: Europe's Gas Pipeline System (Bailey, o.d.)

The annual volume of Russian gas imported by European countries is necessary to calculate the maximum flow realistically. The central data source was EUROSTAT (2023), which provided gas import data for most European nations for the year 2021. Various datapoints that were missing, for example for Switzerland's demand were added with divers' sources (Bocksch, 2021; European Commission, 2022; Verband der Schweizerischen Gasindustrie, 2023). In determining the reference year for the analysis, 2021 was selected because it was the last unaffected year prior to the disruptions following Russia's full-scale invasion of Ukraine, offering a baseline scenario for normal European gas imports. It is, therefore, best suited to be the reference year for a back to "business as usual" scenario in Europe's gas imports. Figure 3 displays the amount of gas imports in bcm for 2021. Germany is by far the biggest importer, followed by Italy. Additionally, other European nations to the west of Germany, such as the Netherlands, France, and Belgium, also demonstrate substantial demand. For the purpose of our analysis, the maximum possible flow is therefore not only compared to Germanys demand for Russian gas but also with the summed-up demand of all nations whose gas has to travel through Germany to reach them. This comparison is crucial because in a situation in which Germany wants to import pre-war volumes of gas from Russia, other European nations are likely to follow. It is therefore a relevant question if their demand could be matched as well.

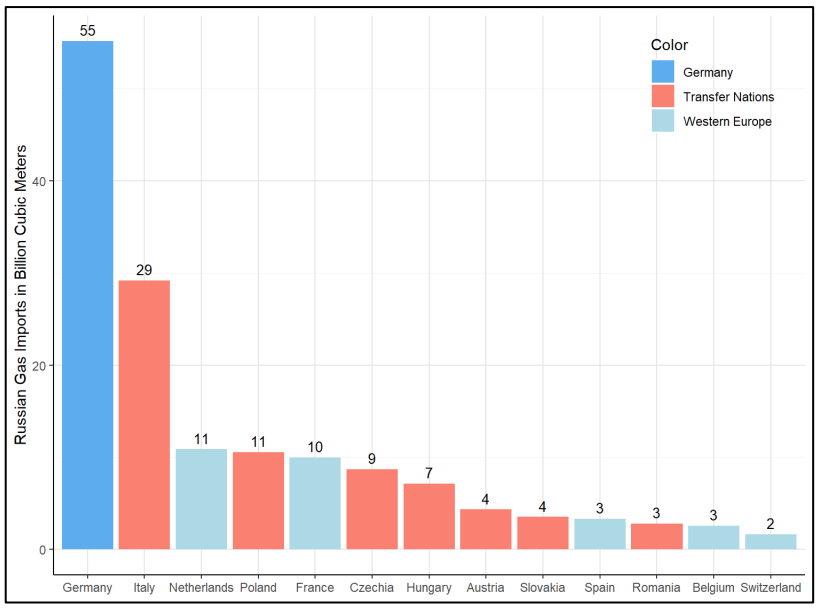


Figure 3: Russian Gas Imports 2021 of chosen European nations

Overall, western Europe's annual gas demand sums up to 83 bcm, significantly exceeding Germany's demand of 55 bcm. It is evident from the data that the demands of transit nations such as Poland, which are expected to resume pre-war import levels under a scenario where Western Europe returns to previous levels, are also substantial. The cumulative demand of all transit countries reaches 67 bcm, including Italy, which is categorized as a transit country because its gas needs are met through Austria via the Trans-Ukraine pipeline, thereby impacting potential inflows to Western Europe (ENTSOG, 2024).

Gas Flow Analysis: Research Design

To test the first three hypotheses, I constructed an oriented graph based on the European gas network data. For this graph, I calculate the maximal physically possible flow between Russia and Germany across various scenarios, representing the different states of the physical infrastructure thematized in the hypotheses. These maximum flows are then compared both among themselves and with the demand of Germany and Western Europe to assess the impact of the Nord Stream attack on the ability to import pre-war levels of gas from Russia in different scenarios. The approach is informed by a study which

employed a similar methodology to explore the effects of pipeline malfunctions on the Chinese gas system's transfer capabilities (Wang, Zhang, Li, Liu & Han. 2020, p.1).

In the oriented graph, each node represents a national gas balance zone, and each orientated edge signifies a directed pipeline connection between balance zones. I selected balance zones as nodes because each functions as a distinct reservoir where supply meets demand, facilitating the allocation of national demand to them (European Commission, 2014, p.17). Therefore, each node incorporates the corresponding national demand as an attribute, essential for the maximum flow analysis and for adjusting the inflow at each node in line with the demand. The edges represent the physically available pipeline connections between balance zones and are oriented according to their flow direction, each with a maximum possible flow attribute.

Subsequently, a maximum flow analysis is conducted on this oriented graph between Russia and Germany. Maximum flow is defined as the greatest physically possible flow between supply and demand, considering pipeline constraints and demand on the way (Wang et al., 2020, p.4). To facilitate this calculation, the graph is transformed into a single-source, single-sink graph, which is shown in Figure 4 (Wang et al., 2020, p.4). Here, Russia serves as the source, and Germany as the sink. The graph is reconfigured to include all national demands en route to Germany, creating a direct linkage between nodes with demand and the German sink as virtual pipelines denoted in orange in Figure 4, where the maximal flow capacity equates to the demand at these nodes. Given the algorithm's preference for shorter paths, the available flow into other connections diminishes due to the actual demand at each node, which is then routed directly to the sink and subtracted from the maximal flow calculated. It is important to note that my analysis does not simulate the pipeline network from Germany to the other western European nations that use Germany as a transfer hub for Russian gas. I assume that the gas network from Germany to other western European consumers has adequate capacity to meet their prewar demand, given that enough gas reaches Germany, as no significant changes have occurred in that system

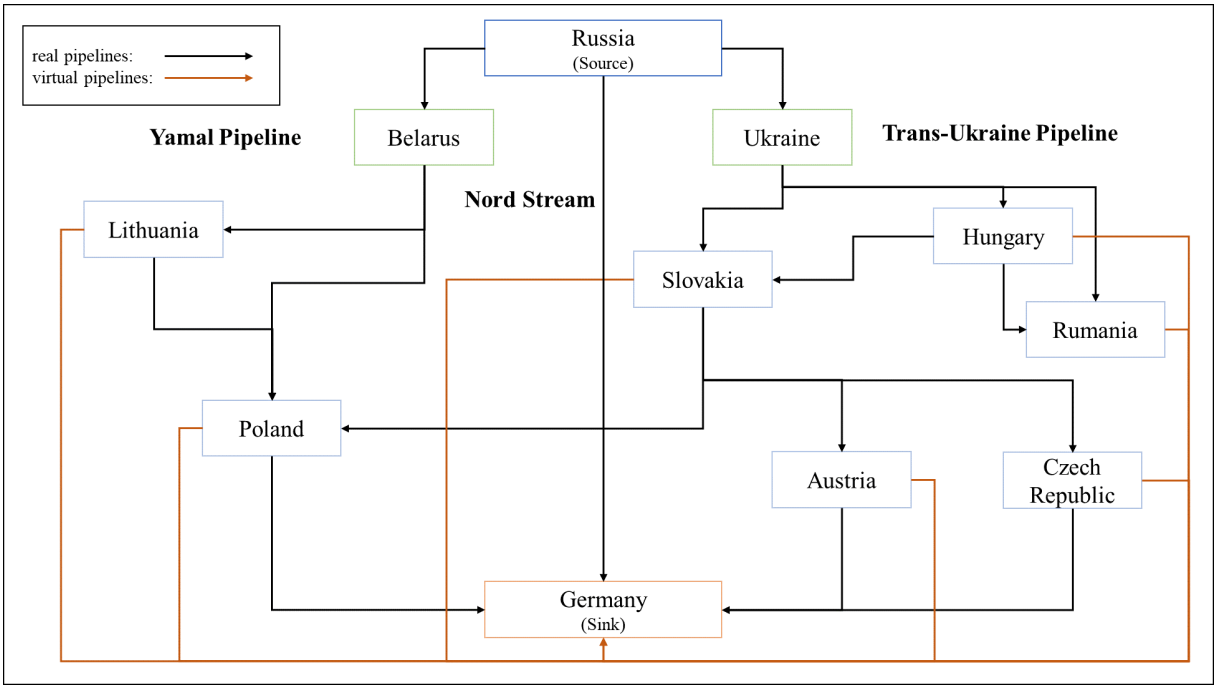


Figure 4: single Source single Sink Graph of Europe's Gas Network

since the reference year of 2021. Additionally, I am not simulating whether sufficient gas could reach Belarus and Ukraine from Russia to maximize the outflow to the EU, due to the lack of available data for those connections and the absence of such a bottleneck in the literature. The foundational paper by Wang et al. utilized a Ford-Fulkerson algorithm to determine the maximal possible flow from one source to one sink in the modified graph (2020, p.5). In my analysis, I employ the "max_flow()" function from the igraph R-package, which implements an algorithm by Goldberg and Tarjan equivalently used to the Ford-Fulkerson method (1988, p.921).

This maximum flow analysis is conducted under various scenarios, whether the Nord Stream attack has occurred or not, each representing a different European infrastructure state. By comparing these flows and analysing the utilization rates of different edges in the system, I can determine the impact of the Nord Stream attack on the maximal possible flow and pinpoint where bottlenecks emerge in different scenarios. The derived maximum possible flow is then compared to the pre-war demand for Russian gas by Germany and Western Europe to assess whether the impact effectively lowers the system's capacity below the demand, or merely reduces surplus capacity and system resilience. These findings enable me to either confirm or refute the hypotheses regarding the impact of the Nord Stream attack on the physical import capabilities of Germany and Western Europe in different scenarios.

Gas Flow Analysis: Results

In my analysis, four different scenarios concerning the state of the gas infrastructure in Europe were discussed to compare the impact of the Nord Stream attack under various assumptions. The first scenario, the pre-war scenario, involves all pipelines, including Nord Stream 1 and 2, that are physically capable of delivering gas. This scenario serves as the baseline. In the second scenario the Nord Stream attack had occurred, and the capacity of Nord Stream is therefore reduced to 27.5 bcm which represents the surviving pipeline's capacity (Reuters, 2022). All other pipelines remain operated in this scenario. This represents the physical capabilities of the infrastructure at that moment, excluding political decisions like the closure of the Yamal Pipeline in 2022 (Strzelecki, 2022) and thereby tests hypothesis 1. The third scenario depicts the condition of Europe's gas infrastructure post the 2022 Yamal Pipeline (Strzelecki, 2022) closure and the Nord Stream attack, and therefore tests hypothesis 2. This scenario details the situation in Europe from the September 2022 Nord Stream attack to the writing of this thesis in Spring 2024. The fourth scenario explores the potential future, representing a scenario where the Trans Ukraine Pipeline is also closed, leaving the surviving Nord Stream pipe as the only connection from Russia to the Western European gas markets. This scenario, which tests hypothesis 3, is expected to unfold at the start of 2025 when the gas transport agreement between Russia and Ukraine, upheld by Ukraine throughout the war, is due to expire (Vakulenko, 2024; Gavin, 2023). In these four scenarios, the maximum possible flow, accounting for pipeline capacities and demand along the route, was calculated and compared to Western Europe’s and Germany's demand. Additionally, each scenario was analysed with a fully operational Nord Stream pipeline to assess the impact of the Nord Stream attack by contrasting scenarios with and without Nord Stream.

	Scenario 1	Scenario 2 (Hypothesis 1)	Scenario 3 (Hypothesis 2)	Scenario 4 (Hypothesis 3)
Maximum Flow	157,5 bcm	91,0 bcm	47,8 bcm	27,5 bcm
Western Europe's demand covered (without NS)	-	109%	57%	33%
Germany's demand covered (without NS)	-	168%	89%	50%
Western Europe's demand covered (with NS)	188%	188%	138%	88%
Germanys demand covered (with NS)	292%	292%	211%	174%

Table 1: Results Maximum Flow analysis

Table 1 outlines the results of this analysis. Scenario 1 reveals substantial overcapacity in the infrastructure prior to the Ukraine war. The maximum flow of 157.5 bcm would have been sufficient to cover 188% of Europe's pre-war demand. These over-capacities resulted from the construction of Nord Stream 1 and 2, which were primarily politically motivated to bypass Ukraine as a transit nation as it was outlaid in the literature analysis (Lang et al., 2016, p. 6). The consequences of this infrastructure duplication are evident in scenario 2, where despite the destruction of 75 percent of the Nord Stream capacities, the physical transfer capacities still fully meet the demand of Western Europe. This leads to the validation of hypothesis 1, that even after the North Stream attack Western Europe would be physically capable to cover its pre-war gas demand. In scenario 3, the closure of the Yamal pipeline, in addition to the Nord Stream attack, leads to major supply shortages. Europe's pre-war demand could now only be met to 57%. If the 43% supply gap were evenly allocated among Western European nations, Germany would encounter a 25.9 bcm shortfall, representing 30% of its 2021 gas imports (Eurostat, 2023). If Nord Stream was still fully operational, the results show that despite the closure of the Yamal pipeline, Western Europe would be able to fully met its demand. The maximum flow analysis confirms therefore hypothesis 2. In scenario 4, with the closure of the Trans Ukraine pipeline, the situation becomes even more difficult for Western Europe. With the surviving Nord Stream 2 pipe, only 33 percent of Western Europe's demand could be met, resulting in a supply shortfall of 55.5 bcm on the Western European gas markets. Even if Germany decided to consume all the incoming gas on its own, it could only cover 50% of the pre-war demand. With a fully operational Nord Stream, however, 88% of Europe's demand could still be met, resulting in only minor supply shortages. This leads me to the acceptance of hypothesis 3.

Overall, it can be concluded that since Europe has been in scenario 3 since the Nord Stream attack and is moving towards scenario 4 with the likely closure of the Trans Ukraine pipeline by end of 2024, the Nord Stream attack has greatly affected Europe's capability to import Russian gas at pre-war levels. With Nord Stream fully operational, most of Western Europe's demand could have been met in all cases, whereas without Nord Stream, the closure of the Yamal pipeline and the potential closing of the Trans

Ukraine impacts Europe's import capabilities significantly. These results lead to the acceptance of the first 3 hypotheses.

However, the impact of this reduction in import capabilities remains unclear, as Europe has not fully utilized its existing import capacities, as shown in Figure 5, which compares the monthly maximum possible inflows to Europe with the actual monthly gas imports from Russia to the EU. It is evident that the existing capabilities were never fully utilized, and the imports declined more rapidly than the capabilities. The drop in Europe's gas imports was therefore not primarily driven by reduced import capabilities, but probably by the political will to decouple Europe from Russian gas exports. The reduction in physical import capabilities can therefore be more seen as a reduction of theoretical options for Western Europe, where it is highly debatable if it would have been utilized if existing.

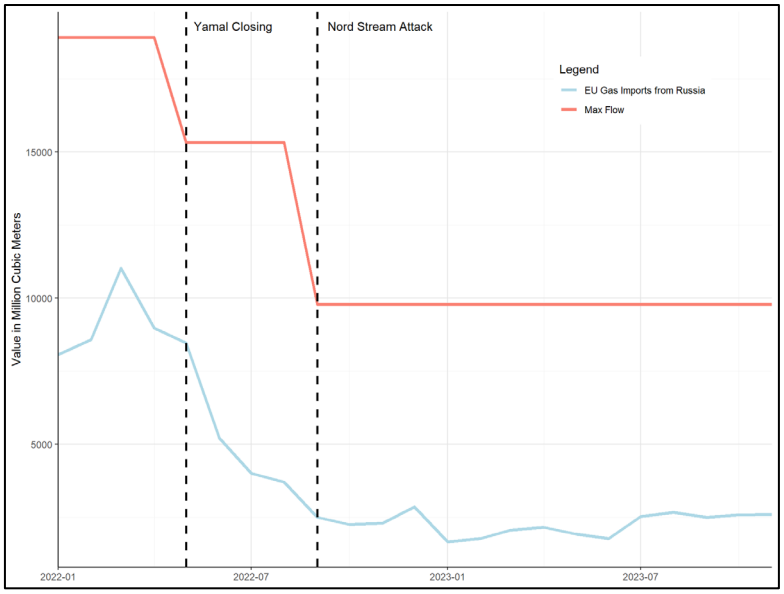


Figure 5: Maximum Flow utilization

Gas Price Analysis

During this part, I intend to elaborate on the gas price analysis that I used to provide insights into the potential price effects of the Nord Stream attack. I shall test hypotheses 4, 5, and 6 by utilizing a multiple linear regression that assesses the impact of various structural factors on the gas price. Initially, I will outline which variables were included, how they relate to the hypotheses, and provide the theoretical reasoning for their inclusion. Afterwards, I plan to explain how the data sources were cleaned and augmented to be usable for the regression analysis. Subsequently, I am going to discuss why I opted to use a multiple linear regression model and how the model tests the hypotheses. Finally, I am set to present the results of my model and interpret them.

Gas Price Analysis: Data

This chapter will give an overview of the variables and data sources included in the model. First of all, I will outline which variables were used based on theoretical arguments, and provide descriptive statistics and data sources for each variable. Afterward, I will explain in which steps the data was cleaned and processed to make it usable for the linear regression model.

Data: Data Sources and Descriptive Statistics

All data sources were gathered for the period from August 2018 to January 2024. I selected this timeframe for analysis because it allowed the collection of all data sources over the longest possible period, thus providing the most extensive database. Including the period before the Nord Stream attack and the beginning of the Ukraine war helps prevent overfitting to the substantial price fluctuations triggered by news post-2022. This method captures therefore more accurately the influence of structural factors on gas prices.

The dependent variable, the gas price, is operationalized using the Dutch TTF natural gas Futures price. Futures are selected for all price operationalizations due to their precision in reflecting prices at specific times. The Dutch TTF price, noted for its high liquidity and the highest churn rate in Europe, serves as the benchmark for European natural gas prices (Heather, 2021, p. 5-6). The data is retrieved from investing.com, which provides historic daily price data from every trading day since 2017 (2024b). Figure 6 illustrates that gas prices remained stable and low until mid-2021, rising sharply with escalating Ukraine-Russia tensions. Following Russia's full-scale invasion in 2022 and subsequent gas disputes in Europe. Afterwards prices fluctuated markedly, peaking before a decline in 2023. This period was overall characterized by significant volatility.

Coal and oil prices serve as control variables, with coal included due to its role as a pivotal gas substitute in electricity generation, where rapid adjustments in power output can buffer renewable fluctuations equivalently to gas. This substitution effect could then have a positive influence on the demand for natural gas when coal prices rise, and indeed numerous papers show a significant positive effect (Halser & Paraschiv, 2023, p. 7-10; Nick & Thoenes, 2014, p. 522; Obadi & Korcek, 2020, p. 323). In Germany, both coal and natural gas are crucial for electricity production, emphasizing the inclusion of coal prices

in the model (Statista, 2023; Statistisches Bundesamt, 2023). The Rotterdam coal futures price is chosen as the operationalization as it is the benchmark for EU coal prices (Sribna et al., 2019, p. 4). Figure 6 reveals that coal prices, though similar in trend and volatility to gas, maintained higher price levels after the initial price shock faded out. The data is retrieved from Investing.com (2024c)

Although literature often positions oil as a gas substitute influencing prices (Halser & Paraschiv, 2023, p. 23; Nick & Thoenes, 2014, p. 520; Obadi & Korcek, 2020, p. 323), its role in German electricity production is minimal, casting doubt on its impact (Halser & Paraschiv, 2023, p. 23; Statistisches Bundesamt, 2023). Nonetheless, considering significant historical correlations (Brown & Yttcel, 2008, p. 59) and recent structural changes post-Ukraine invasion which could impact the relationship, I chose to include the oil price as a control variable. Brent oil, as the principal reference for European oil prices, was selected as the operationalization (Coronado et al., 2017, p. 982). The data source is investing.com (2024a). In Figure 6, it can be seen that the oil price rose equivalently to the gas and coal prices in 2021, but the ascent was less steep. It is also less volatile in the observed timeframe and recovers the least compared to pre-war levels.

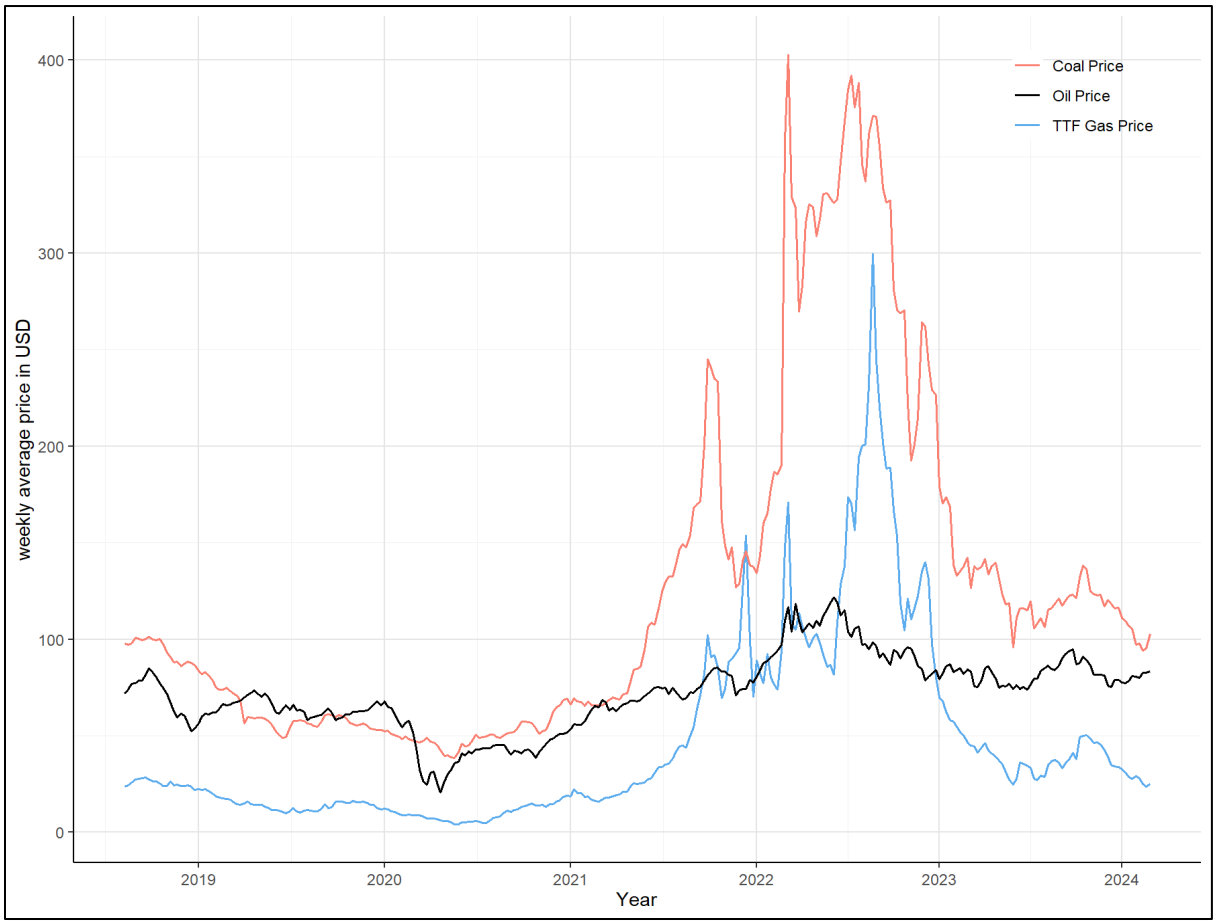


Figure 6: Gas-, Coal- and Oil price compared

To test Hypothesis 4, which states that the reduction in gas supply caused by the Nord Stream attack led to higher gas prices, I included the total EU gas inflow as an independent variable. This variable was found to have an influence on the gas price in times of supply disruptions in other cases (Nick & Thones, 2014, p.522). The logic is that a reduced supply meets a similar demand, which then drives up the prices. Gas inflow data was collected from an aggregated dataset (Zachmann, William, Keliauskaite &

Sgaravatti, 2024) based on ENTSOG data (2024). ENTSOG tracks all inter-balance zone gas movements in Europe, and the aggregated dataset has structured this data in a form that presents how much gas from which sources was imported into the EU at a given point in time. Figure 7 shows the development of the weekly total gas inflow into the EU with the blue line. It can be seen that there were fluctuations, but until 2022 the total gas imports always moved around the mark of 1000 mcm per week. Then within 2022 the total gas inflow dropped and has fluctuated since then around the mark of 800 mcm per week.

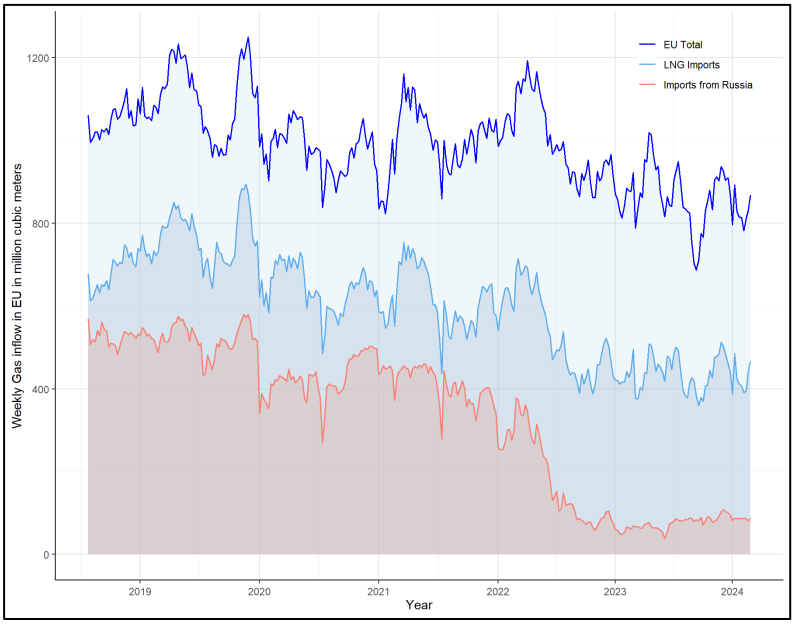


Figure 7: weekly Gas inflow too the EU

The total gas imports of the EU from Russia and the total LNG inflows were also included as independent variables to test hypothesis 5, which discusses the potential effect of changes in the gas supply sources on the gas price. It can be assumed that a higher share of LNG on the total gas supply could increase the gas price as a consequence of higher transportation costs (Molnar, 2022, p.54-56). The role of LNG imports in the European gas market was already previously explored in the literature, and a small significant negative effect is found. This effect is attributed to the consequence that an increase of LNG imports increases the total supply on the European gas market which could then drive down prices (Nick & Thoenes, 2014, p.523; Obadi & Korcek, 2020, p.6). However, this rationale does not hold post full-scale invasion because, as it can be seen in Figure 7, which displays in the coloured surfaces the share of LNG and Russian imports on the total EU imports, increased LNG imports compensated the decrease of Russian pipeline imports after 2022, without boosting overall supply. Assessing the impact of substantially increased LNG imports on the gas price is therefore of general interest in addition to its relevance for the assessment of hypothesis 5. Data on LNG and Russian gas imports were sourced from an aggregated dataset based on ENTSOG records (Zachmann et al., 2024).

A dummy variable was introduced to directly assess the Nord Stream attack's impact on gas prices, independent of gas inflow or source changes, and therefore operationalize hypothesis 6. It is set to zero before and one after the attack. The variable enables the isolation and analysis of the direct price impacts attributable solely to the Nord Stream attack, unexplained by other model variables.

The number of heating degree days, used to quantify necessary heating, was found to have a significant impact on gas demand in various studies (Halser & Paraschiv, 2023, p.23; Nick & Thoenes, 2014, p.522) which makes it crucial to include that factor as a control variable. Heating degree days operationalize necessary heating by counting the degrees that the daily average temperature is under 18 degrees (Halser

& Paraschiv, 2023, p.12). Official statistics on Germany's heating degree days are available only up to the end of 2022 (Eurostat, 2023). I computed therefore the amount of heating degree days using Basel's average temperature data, sourced from Metroblue (2024), which reliably approximates Germany's heating degree days with a historical correlation of 0.98. To account for seasonal variation in the data and only include unseasonal changes in heating demand I used the deviation of heating degree days from the 3-year average as the operationalization in my model.

Another potential impact factor in the gas demand is industrial production, which accounts for a share of 35% of the total German gas demand in 2022 (Bundesamt für Wirtschaft und Klimaschutz, 2024). Changes in industrial demand for gas could therefore affect the gas price on the demand side, and discussing the impact is especially interesting because in the observed timeframe significant efforts were made to save gas in industrial processes (Bundesamt für Wirtschaft und Klimaschutz, 2022). I therefore include the monthly German industrial production index from the Stats Office of Germany (Statistisches Bundesamt, 2024) as a control variable in my analysis. Literature discussing the German case has also operationalized the industrial gas demand with the industrial index, but found no significant relationship (Halser & Paraschiv, 2023, p.23). However, this conclusion may not apply to the current period, due to the extraordinary surroundings. One challenge that has to be kept in mind while interpreting the results of this variable is potential endogeneity, because the gas price could also affected industrial demand.

Concerning the resilience of the gas supply system, the filling level of gas storage is another potential impact factor on the gas price. Theoretically, higher storage levels provide more flexibility to mitigate unexpected demand surges or supply deficits, thereby helping to stabilize prices. This effect was also found in the literature for the European case (Nick & Thoenes, 2014, p.522; Obadi & Korcek, 2020, p.6). Given the significant fluctuations in storage levels, particularly the historically low levels pre-war, it is crucial to include this variable for a comprehensive analysis. The data is retrieved from Gas Infrastructure Europe (2024). Due to the pronounced seasonal influence on the data, I adjust for seasonality by using the deviation from a 3-year average to focus only on unseasonal variations. Figure 8 displays the deviation from the 3-year average storage volume. Especially surrounding the beginning of the Ukraine crisis in 2022, huge movements happened. From a very high storage volume in the middle of 2020, the storage volumes dropped drastically until the beginning of 2022, when the storage was then rapidly refilled at a historically high level. These huge movements

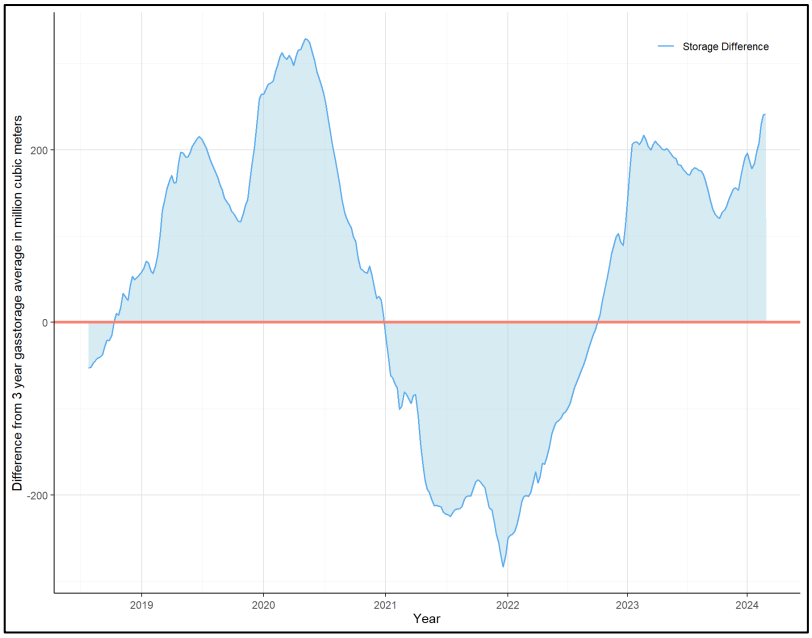


Figure 8: development of the difference from 3-year average gas storage filling

could have affected the gas price, which emphasizes the importance of accounting for storage volumes in the model.

Renewable electricity feed-in has recently been highlighted as a potential determinant of gas prices due to gas's role in stabilizing fluctuations in renewable energy supply. Gas demand for electricity production could therefore be directly connected with the feed-in of renewable electricity, which could then impact the gas prices. Especially in the case of Germany, this could be a huge impact factor since Germany produces around 50% of its electricity demand from renewable sources (Statistisches Bundesamt, 2023). This effect was indeed shown for daily gas price development in a recent paper for Germany (Halser & Paraschiv, 2023, p.3). Since renewable electricity infeed shows some seasonal patterns, with higher feed-ins in the summer, I decided to incorporate the difference from the 3-year average infeed as a control variable in the model. By doing this, seasonality's are filtered out and only unexpectedly high or low infeed have an impact on the model. As data sources, I use Netztransparent (2024), who provide accurate infeed data which I accumulated and processed to get the difference from the 3-year average infeed in GW/h.

These are all variables that were identified based on the literature as having a potential impact on the gas price and are therefore included in the model. For the hypothesis testing, the relevant independent variables are the total gas inflow to the EU for hypothesis 4, and the LNG inflow and Russian imports for hypothesis 5. The dummy variable will analyse the overall impact of the Nord Stream attack and test therefore hypothesis 6. In the next section, I will detail the data cleaning and preparation processes used to estimate the linear regression model.

Data: Cleaning and Processing

In this part, I will outline the steps in which the data was cleaned and adapted in order to be usable in the linear regression model. First of all, the data was aggregated to weekly data, to filter out daily fluctuation noise, and get more of the structural impacts in the model. In the case of the price variables, the industrial index, and the storage difference, an average was formed over the week and for all other variables the sum.

In the next step, the stationarity of the data is tested, which is especially important when using time series data (Halser & Paraschiv, 2023, p.18). I used an Augmented Dickey-Fuller test to control the stationarity, which revealed that all price-, the storage difference-, the Russian gas import-, and the LNG Import data show signs of non-stationarity. To resolve non-stationarity, the data series were differentiated, achieving stationarity as detailed in Appendix 1. This differentiation must be kept in mind during the later analysis. Subsequent testing for multicollinearity among all variables included in the model indicated no significant issues (Appendix 1).

Another factor that has to be considered is the problem of potential endogeneity of the coal price in relation to the gas price. Post-Ukraine invasion, it was hypothesized that increased coal usage in power generation, driven by high gas prices, influenced coal pricing (Sturm, 2022, p.854). Support for this hypothesis is found in the data showing a 9% decrease in gas use and an 8% increase in coal use for electricity in Germany (Statistisches Bundesamt, 2023). On the other hand, there was also a supply

shortage for coal, following the EU ban on Russian coal imports, which was previously a central supplier (Katarzyna & Grudzinski, 2022, p.31). How huge the impact of the ban on Russian coal was, is however highly debated because alternative suppliers like Australia were to a certain extent available on the global market (Katarzyna & Grudzinski, 2022, p.42; Sturm, 2022, p.866). I address this problem of potential endogeneity by lagging the coal price for one period, which should filter out the potential endogenous effect within the same period (Wang & Bellemare, 2020, p.24).

In a final step, I decided to eliminate 3 value points directly after the beginning of the Ukraine war out of the data because an examination with Cook's distance revealed that they have a disproportionately large impact on the fitting of the model (Appendix 1). These points were primarily influenced by news events rather than underlying economic changes, risking model overfitting to transient noise rather than to structural shifts if the points were included.

The cleaned dataset now includes 290 weekly observations from August 2018 to January 2024. All the price data, Russian and LNG import data, and storage differences are differentiated. Additionally, coal prices are lagged by one period. With this cleaned data, the model can be estimated. In the next chapter, I will explain with which method the model is calculated and how it is configured.

Gas Price Analysis: Research Design

I have utilized an OLS multiple linear regression model to estimate the impact of the previously discussed variables in the data part on the dependent variable of the TTF Dutch gas price. I decided to use a multiple linear regression model because it is a common econometric method to estimate the impact of different factors on an independent variable (Anghelache, 2014, p.121), and it has proven to lead to solid results in recent papers discussing the impact of different factors on the gas price (Halser & Paraschiv, 2023, p.23; Obadi & Korcek, 2020, p.322). An alternative approach, which was also used in literature discussing the gas price, is the vector autoregression method (Nick & Thoenes, 2014, p.520; Roberts, 2019, p.17). I opted against the vector autoregression method, typically applied to transient external impacts (Nick & Thoenes, 2014, p.524-526), because the Nord Stream attack fundamentally altered structural factors permanently. Thus, a multiple linear regression model was chosen for its straightforward ability to delineate the magnitude and direction of impact from independent variables on the dependent variable. However, employing a vector autoregression model could offer additional insights, making it a worthwhile extension for future research. The general model which is used during this analysis is outlined in Table 2.

$$\Delta PTTF_t = \beta_0 + \beta_1 EG + \beta_2 \Delta RG + \beta_3 \Delta LNG + \beta_4 \Delta PC + \beta_5 \Delta PO + \beta_6 \Delta SD + \beta_7 HDD + \beta_8 RE + \beta_9 II + \beta_{10} ND + \varepsilon$$

EG = total EU gas inflow	ΔSD = differenced gas storage
ΔRG = differenced gas inflow from Russia	HDD = heating degree days
ΔLNG = differenced LNG inflow	RE = renewable electricity infeed
ΔPC = differenced lagged coal price	II = industrial index
ΔPO = differenced oil price	ND = north stream dummy

Table 2: general multiple regression model

Hypothesis 4, which argued that a decreased gas supply as a result of the Nord Stream attack should drive up gas prices, is tested with the variable differenced EU gas inflows. Following the reduction in gas inflows due to the Nord Stream attack, as revealed by the previous gas flow analysis, this analysis examines the consequent effects on gas prices. If a negative coefficient is found for the differenced EU gas inflow, this would mean that a decrease in the gas inflow would increase the price, which would then lead to the acceptance of the hypothesis. The variables, differenced LNG inflows and differenced gas inflows from Russia, test hypothesis 5, that the source of gas matters for the gas price. If the hypothesis is correct, then the coefficient for the differenced LNG inflow should be positive and the coefficient for the differenced Russian gas inflow negative. This would mean that a higher LNG inflow increases the prices, and a higher Russian gas inflow decreases the prices. The Additional control variables are included to examine alternative price determinants and avoid a omitted variable bias, enhancing the overall understanding of gas price dynamics, with a specific dummy variable isolating the distinct impact of the Nord Stream attack.

Gas Price Analysis: Results

In this chapter, I will examine the results of the multiple linear regression analysis. Therefore, I will first present in which different set ups the model was estimated and explain the rationale for this procedure. Afterward, I will give an overview of the estimated coefficients. These will then be more closely discussed and interpreted in the following chapters.

Table 3 displays the results of the four different regression models which I estimated. In the first model, I only included the dummy variable, operationalizing the North Stream attack, to estimate how huge the explanatory power of a model only including the attack is, and to get an understanding of the scale of the impact of the attack. In the second model, I additionally included all variables on which the North Stream attack had an impact, and which operationalize hypotheses 4 and 5. This model includes all for the hypothesis testing relevant variables. In the next step, I include all non-price control variables which are the amount of heating days, the renewable energy infeed, and the industrial production index. In the final model, I add the differenced gas, coal, and oil price. Overall, the model's explanatory power, quantified by the adjusted R squared, improves with each inclusion of additional variables, underlining that indeed all factors are somehow relevant for the development of the gas price. Nevertheless, even in the final model, the R squared is relatively small with a p value of 0,111, which means that only 11% of the variation in changes in the gas price can be explained with the model. The conclusion based on this is, that even though the fundamental factors included have an impact, unobserved factors, for example news and political developments, played a bigger role in the development of the gas price in the analysed period. Nevertheless, the model provides valuable insight in the role of various factors, which I will interpret more closely in the following chapters.

Concerning the robustness of the model and the results, no problems are found. To address the issue of potential heteroscedasticity in the variance of the error term, robust standard errors were used to calculate the displayed t-tests and corresponding p values (Eugster, 2024, p.18). The independence of residuals is tested with a Durbin-Watson test and more closely examined with an AFC function, and no issues are found (Appendix 2). Also, the normal distribution of residuals is given (Appendix 2), so that

it can be overall concluded that the results of the model are robust since all necessary assumptions for a multiple linear regression hold. There is also no problematic level of multicollinearity found in the independent variables (Appendix 2).

Regression Results

	Dependent variable:			
		`differenced TTF gas price`		
	(1)	(2)	(3)	(4)
Total EU Gas Inflow		-0.011	-0.010	-0.013
		p = 0.123	p = 0.143	p = 0.185
differenced gas inflow from Russia		0.031*	0.026	0.033*
		p = 0.099	p = 0.158	p = 0.094
differenced LNG Inflow		0.037	0.027	0.026
		p = 0.122	p = 0.207	p = 0.198
differenced lagged coal price				0.111***
				p = 0.004
differenced oil price				-0.206
				p = 0.367
differenced gas storage				0.084
				p = 0.407
heating degree days			0.651**	0.711*
			p = 0.020	p = 0.074
renewable electricity infeed			0.024	0.017
			p = 0.387	p = 0.502
Industrial Production Index			-0.074	-0.078
			p = 0.116	p = 0.102
Dummy North Stream attack	-3.003***	-4.783**	-4.168**	-4.507*
	p = 0.009	p = 0.016	p = 0.031	p = 0.055
Constant	0.823	12.250	18.824**	22.372*
	p = 0.245	p = 0.118	p = 0.036	p = 0.088
R-squared	0.0179	0.0367	0.0657	0.104
Observations	290	290	290	290
R2	0.018	0.037	0.066	0.104
Adjusted R2	0.014	0.023	0.043	0.072
Residual Std. Error	9.774 (df = 288)	9.731 (df = 285)	9.634 (df = 282)	9.485 (df = 279)
F Statistic	5.247** (df = 1; 288)	2.715** (df = 4; 285)	2.833*** (df = 7; 282)	3.241**** (df = 10; 279)
Note:				*p<0.1; **p<0.05; ***p<0.01

Table 3: multiple linear regression results

Results: Hypothesis 4

Hypothesis 4 which posits that the Nord Stream attack increased gas prices by reducing the overall supply in Europe's gas markets, was operationalized with the variable of the total gas inflow in the EU. The coefficient for the variable representing the total gas inflow to the EU is consistently negative and small across all models and is insignificant with p-values exceeding 0.1. This indicates that the null hypothesis regarding its impact on gas prices cannot be rejected.

This insignificance is attributed to the major price increase occurring well before the actual decline in supply, visible in Figure 9, likely due to market anticipation of a restricted supply following the onset of the Ukraine war for political reasons. As the supply dropped in the second quarter of 2022, a part of this was already priced in, but still, the prices spiked during this time, explaining why a small effect is

indeed found in the model. However, in the latter half of 2022, prices plummeted despite continued reductions in supply, which can be explained by a concurrent decrease in demand due to significant gas savings across all sectors. The EU exceeded its gas savings goal of 15% achieving 18% savings (European Commission, 2024), and savings were even higher in Germany with almost 20% (Ruhnau, Stiewe, Muessel & Hirth, 2023, p. 624-625). The approximately 15% lower total gas inflows (Zachman et al., 2024) were therefore met with even higher savings in all sectors, which countered the market anticipated supply-demand gap, and led to dropping prices despite an overall reduction in gas supply. The reduction in gas supply, evident in the data and partly due to Western Europe's inability to import pre-war levels of Russian gas following the Nord Stream attack, did not lead to systematically higher prices but instead to widespread gas savings and reduced demand. Consequently, Hypothesis 4 must be rejected as no significant impact from the operationalized gas supply metric was observed.

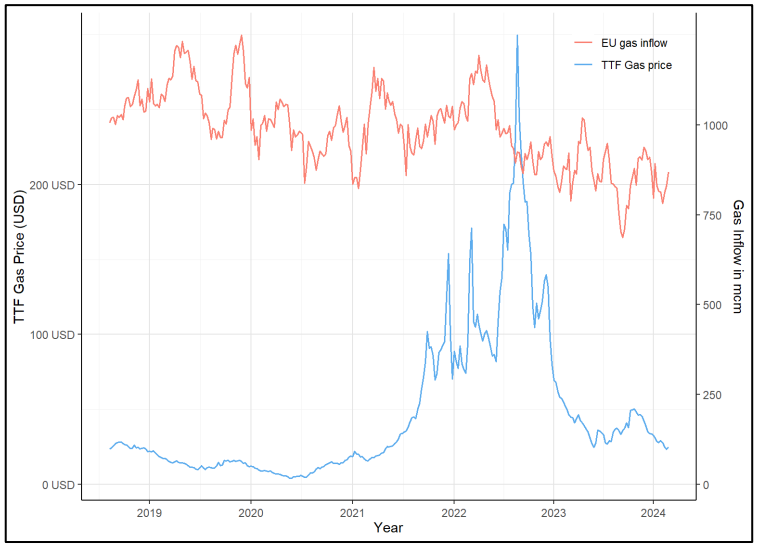


Figure 9: development gas price and total EU gas inflow

Results: Hypothesis 5

The variables of the difference in LNG inflow and the difference in Russian gas exports to the EU were used to test Hypothesis 5. Hypothesis 5 stated that Nord Stream raised the gas prices because a higher amount of gas has to be imported with LNG, which faces higher transportation costs than pipeline gas. If this hypothesis were true, a positive impact of LNG imports on the gas price, and a negative impact of Russian pipeline gas inflows would have been expected. However, in our analysis, the coefficient for both variables is consistently positive. Notably, the Russian gas inflow shows significance only in the second and fourth models, reaching a marginal significance level of 0.1. The difference in LNG inflow does not have a significant impact in any model. Since the Russian coefficient is not as expected negative, and the LNG inflow is positive but insignificant, the hypothesis that the Nord Stream attack had an impact on the gas price by changing the source of supply, must be denied since no impact of the source of gas on the gas price has been found.

This result is remarkable considering prior studies identified a negative impact of increased LNG inflow on European gas prices, attributed primarily to an increase in supply (Nick & Thoenes, 2014, p.523; Obadi & Korcek, 2020, p.6). The absence of this relationship in our models may indicate that the earlier explanation, that during the highly relevant period after 2022 LNG did not represent an additional source but rather compensated for reduced pipeline supply, is accurate. The fact that the higher transportation costs of LNG, relative to pipeline gas, do not result in higher prices might stem from the relatively modest transportation costs of $3-14 per MW/h from the US to Germany (Molnar, 2022, p.55),

especially when compared to the observed gas prices. At peak pricing times, with gas prices around $250 per MW/h, these transportation costs become negligible, with other factors predominating in their influence on price. However, when gas prices fall, as previously seen, the proportion of transportation costs in the total cost increases, potentially making these costs significant in price formation. This development has to be observed in the future.

Results: Hypothesis 6

Hypothesis 6 utilized a dummy variable that shifts to one following the Nord Stream attack to measure the attack's impact through the variable's coefficient. Given the negative and significant coefficient in all models, the null hypothesis is rejected, demonstrating a negative effect on gas price differentials. Consequently, I decisively reject Hypothesis 6, as it required a positive effect from the dummy variable on gas prices to be validated.

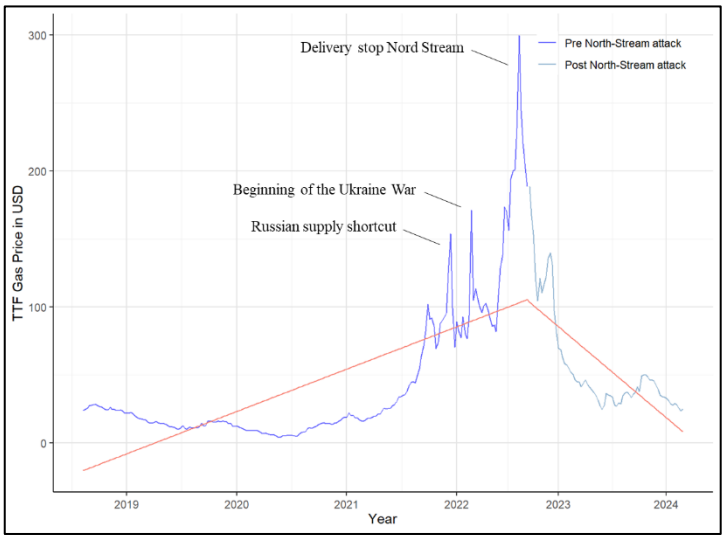


Figure 10: development of the gas price over time

The negative coefficient is illustrated in Figure 10, where it is evident that price spikes occurred prior to the attack and subsided afterwards, as depicted by the red segmented regression lines. This trend might result from Russia's earlier reduction and eventual cessation of gas deliveries through Nord Stream 1, a full month before the attack (RND, 2023). Furthermore, the German government halted the commissioning of Nord Stream 2 immediately after the onset of the full-scale invasion of Ukraine, reflecting a broader European and German strategy to disengage from Russian gas well before the attack (Jong, 2023, p.9). Additionally, past actions by Russia, such as reducing supply amidst high prices in December 2021, might have cast doubt on the reliability of its gas deliveries (Strachota, 2021). All these factors likely led the market participants to the conclusion that in the near future Nord Stream 1 and 2 would not be utilized, which led to the price spikes during the time of these events which are visible in Figure 10. Subsequently, the actual destruction of Nord Stream became irrelevant to market dynamics, as its capacities were already deemed unusable due to prior geopolitical circumstances and decisions. Thus, Nord Stream was effectively rendered obsolete politically before its physical destruction, with the loss of its supply capacities already factored into market prices at the time of its demise.

Interpretation Control Variables

In this section, I will explore how control variables provide alternative explanations for the exceptionally high gas prices during the observed period and evaluate a range of structural factors influencing gas prices. I will start with the impact that potential subsidies for gas in electricity production had. What sticks out is the highly significant impact that the lagged coal price had in my model. A positive coefficient indicates that rises in coal prices positively affected gas prices in the following week, corroborating the initial theory of a positive correlation. This result underlines that the simultaneously occurring shortfall in the coal supply and the corresponding increase in the coal price, caused by the import ban of Russian coal from the EU (Katarzyna & Grudzinski, 2022, p.31), worsened the gas price crisis by making it more expensive to substitute gas in electricity production with coal. These results highlight the necessity of considering fuel subsidies in discussions on energy dependency and security. While Europe's reliance on Russian gas was already a prominent topic, the large volumes of coal imports, which further aggravated the gas supply's vulnerability, lacked similar scrutiny. Oil, as a potential additional subsidy to gas, is not found to have a significant impact on the gas price as expected and already shown in previous literature (Halser & Paraschiv, 2023, p. 23). Consequently, these results endorse the view that oil does not serve as a viable alternative to gas in the German and European markets. The renewable energy infeed to the electricity net has also no significant impact on the gas price, despite previous literature indicating a connection (Halser & Paraschiv, 2023, p.23). The fact that the major difference lies in the timeframe, my model uses weekly data and the model of Halser & Paraschiv daily data (2023, p.9), points to the conclusion that the impact of renewable electricity infeed on the gas price is only short-lived and occurs therefore only in daily data.

Now I want to take a closer look at variables impacting the demand for natural gas. An examination of variables affecting natural gas demand reveals that deviations from the average heating degree days positively and significantly impact gas prices in my model, confirming theoretical expectations that increased heating demand drives up gas prices. While this analysis sheds light on a contributing factor to gas prices, it does not account for the extreme price levels observed in 2022. Notably, 2022 experienced 38 fewer heating degree days compared to the prior three-year average, suggesting reduced heating requirements due to warmer conditions. This factor, therefore, had a moderating effect on the development of gas prices during the crisis. The Industrial Index shows no significant correlation with gas prices in my analysis, an unexpected finding given the substantial role of industrial gas savings in reducing demand in Germany (Ruhnau et al., 2023, p.623). In the short term, these gas savings appear to have no negative impact on industrial production, with the median industrial production index remaining stable in 2022 compared to 2021, explaining why the index's evolution did not affect gas prices. However, it remains questionable whether this can be sustained in the long term, as the index declined in 2023, potentially indicating that the savings achieved without reducing production were only feasible for a limited time, perhaps by utilizing reserves of gas-intensive products. Future observations are necessary to fully understand these dynamics; however, for the timeframe under review, industrial production, as measured by the industrial production index, did not drive gas prices, as the demand for gas within the industry was decoupled from its production levels.

The last discussed factor, which could have an impact on the gas price and was indeed heavily debated in public discussions, is the fillings of the gas storage. My model finds here for the variable changes in the deviation from the 3-year filling average, operationa-lizing this factor, a positive coefficient which is insignificant. This finding is unexpected, as earlier research has consistently shown that increased gas storage levels tend to lower gas prices (Nick & Thoenes, 2014, p.522; Obadi & Korcek, 2020, p.6). The reason for this change in relationship can be traced back to the developments in gas storage levels. As clearly shown in Figure 11, the amount of gas stored was significantly below average at the start of the sharp price increases in the fourth quarter of 2021. When the Ukraine war began, and with it concerns over supply security, gas storages were rapidly filled throughout 2022 during a period of peak gas prices. The swift increase in gas storage, amid escalating prices and limited supply, generated extra demand, exacerbating price rises instead of enhancing market stability and resilience. This explains the absence of the expected price-dampening effect of increased storage levels, revealing instead a non-significant, albeit positive, influence on prices.

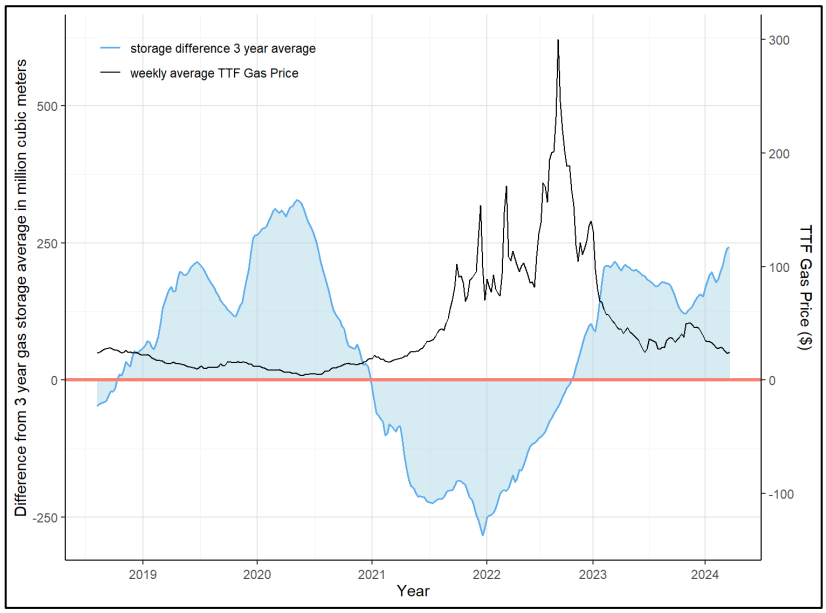


Figure 11: gas storage volumes and gas price compared

Discussion

I answered the general research question, "What impact did the Nord Stream attack have on the European gas market?" in two steps. In the first step, I analysed the impact of the attack on the Russian gas import capabilities of western Europe in a maximum flow analysis. Afterward, the impact of the attack on the gas price was analysed.

Regarding the impact of the Nord Stream attack on the gas import capabilities, a significant impact was found in the maximum flow analysis. Based on the existing infrastructure's physical capabilities, sufficient pipeline capacities remained to import pre-war gas volumes after the Nord Stream attack, confirming Hypothesis 1. This finding supports the analysis in the literature that Nord Stream 1 and 2 were not constructed because additional transport capacity was needed to meet Europe's demand for Russian gas, but rather to bypass transit nations like Ukraine and Poland (Lochner & Bothe, 2007, p. 15). However, in the post-2022 scenario, in which the Yamal pipeline through Poland is closed (Strzelecki, 2022), the Nord Stream attack led to a situation in which only 57% of western Europe's demand could be met, proving Hypothesis 2 that pre-war import volumes were unattainable for western Europe in such a situation. This worsens in a potential future scenario, in which Ukraine also closes the Trans-Ukraine pipeline. In that scenario, only 33% of western Europe's demand can be met, leading to the acceptance of Hypothesis 3. In summary, the Nord Stream attack had indeed a significant impact on western Europe's ability to import pre-war volumes of gas from Russia in the situation that it faced since 2022. However, due to the geopolitical environment, this has only mattered to a limited extent because the import volumes were decreased to even lower levels due to political reasons. The destruction of Nord Stream had, therefore, no practical impact on the gas supplied by Russia, since import volumes over pipelines were constantly lower than physically possible but must be viewed as a destruction of a theoretically available option, forcing western Europe to stick to the gas decoupling path. Without the Nord Stream attack, western Europe and Germany could have returned to nearly pre-war gas import volumes from Russia, regardless of the political situation of the land pipeline connections. Evaluating whether this would have been utilized or impacted politics is beyond the scope of my paper and a topic for further research, which is especially intriguing given that some German political parties have advocated for restoring gas relationships with Russia and repairing Nord Stream 1 and 2 (Bundestag, 2023), and the Kremlin offered to restore gas exports (Zeit Online, 2023).

Despite demonstrating the significance of the Nord Stream attack on import capabilities, my analysis cannot find a significant impact on gas prices. Hypothesis 4 must be rejected, as no significant relationship was found between total EU gas imports and the gas price, indicating that reduced gas inflow capabilities due to Nord Stream did not significantly impact the gas price. The cause of this is mainly that the price went up before the actual drop in demand, anticipating the politically caused decoupling of Russian gas, and that the drop in demand was met with decisive gas savings across all sectors. As a result, when gas supply declined due to decreasing Russian imports, the supply-demand gap was less pronounced than the markets anticipated, leading to declining gas prices. However, these findings cannot be transferred to potential future attacks on energy infrastructure, since in that case, the drop in supply would come totally unexpected. The finding that the reduced supply led to no price impact in my paper cannot be applied to potential future attacks on fully operational energy infrastructure. My

model also found no significant price impact of the source of gas, leading to the conclusion that rising transport costs due to increased LNG imports were too minor to have a notable impact on the gas price in the observed period. That no negative effect was found, in contrast to the findings of previous literature (Nick & Thoenes, 2014, p.523; Obadi & Korcek, 2020, p.6), underlines that the structure of the market has changed. LNG is no longer an additional supply that reduces prices but rather necessary to fulfil the baseline demand due to reduced pipeline imports. Thus, Nord Stream did not affect gas prices through a shift in supply sources, necessitating the rejection of Hypothesis 5.

Finally, the total effect, estimated through a dummy variable, indicated a negative price impact of the Nord Stream attack, leading me to reject Hypothesis 6 and conclude that the destruction of Nord Stream did not drive-up gas prices in the observed period. Gas prices peaked due to anticipated supply drops arising from the geopolitical situation, rather than from any actual physical inability to import pre-war gas levels from Russia. I can therefore not find an immediate impact of the Nord Stream attack on the gas price. However, the economic consequences of the extensive gas savings required across all sectors remain uncertain. From the fact that that the industrial index has not dropped despite the huge savings (Statistisches Bundesamt, 2024), I concluded that the gas savings were possible decoupled from reduced industrial production in the short term. However, signs suggest this might not be sustainable in the long run, and further research is needed to assess the economic consequences of the required gas-saving program in Germany and the EU. These economic consequences, when assessed more closely, could be the real economic consequences, of the Nord Stream attack enforced decision to decouple from Russian gas and reduce gas demand.

Concerning the gas price model tested control variables, the significant positive impact of the coal price stood out. The simultaneous scarcity of coal, in addition to gas, because it was also heavily imported from Russia before the Ukraine war (Katarzyna & Grudzinski, 2022, p. 31), made gas substitution in electricity production challenging, explaining why the electricity sector lags behind in gas savings compared to heating and industry (Mannhardt, Gabrielli, & Sansavini, 2023, p. 8). This result supports previous literature, which identified the coal price as an important driver of the gas price. (Halser & Paraschiv, 2023, p. 7-10; Nick & Thoenes, 2014, p. 522; Obadi & Korcek, 2020, p. 323). The high relevance of coal prices emphasizes the need to consider the availability of potential subsidies and their correlated dependencies in supply resilience analysis, often overlooked in previous studies (Sesini, Giarola & Hawkes, 2020, p. 9). What is surprising is the insignificant impact of the gas storage volumes on the gas price, despite earlier findings (Nick & Thoenes, 2014, p.522; Obadi & Korcek, 2020, p.6), suggesting that the filling (of the gas storages in a situation of already tight supply did not contribute towards decreasing prices via increased resilience because the additional demand drove up prices on its own. This inability to utilize available gas storages to buffer price spikes warrants further analysis to better understand their impact on the market during crises and to enhance their future utilization.

The primary limitation of my research is the observed timeframe. Less than two years of data were available post-Nord Stream attack, which means the long-term structural effects might not have fully manifested when this thesis was written. The results of this study should therefore be viewed as a first estimation of the consequences in the two years following the attack and not as a final conclusion. Additionally, my use of multiple linear regression for price estimation meant I could not capture

dynamic relationships among the independent variables, which should be considered when interpreting the results. A vector autoregression model could offer additional insights into the dynamics of the system and therefore be a valuable addition to my model, which focused more on the structural impacts (Gottschalk, 2001, pp. 17-19). Finally, my paper is clearly limited in scope, as it solely focused on maximum transport capacity and price impact. The overall economic consequences of the necessary gas savings and the political implications of the attack's impact on reducing import options are not discussed, though they could be major additional impacts of Nord Stream. Building on this paper's fundamental findings about the direct impact on the European gas market, further research is necessary to fully understand the impact, consequences, and costs of the Nord Stream attack for Europe.

Conclusion

In this paper, I have examined the impact of the Nord Stream attack, which took place in September 2022, damaging three out of four pipes of Nord Stream 1 and 2, on the German and western European gas markets. I have focused and structured my research approach on two key aspects: the impact of the attack on western Europe's gas import capabilities from Russia over pipelines and the effect on gas prices. The impact on import capabilities was examined through a maximum flow analysis based on a paper by Wang et al. (2020). I used a multiple linear regression model to assess the impact of different structural factors on the gas price. By including a dummy variable representing the Nord Stream attack, along with various influenced variables, this approach enabled me to estimate the effect of the Nord Stream attack on gas prices.

Through this analysis, I derived three key findings. First of all, concerning import capabilities, the Nord Stream attack effectively eliminated western Europe's option to import pre-war volumes of Russian gas following the closure of the Yamal pipeline in April 2022. However, this reduction in import capabilities has not actively reduced import volumes, as they have consistently remained below the maximum import capabilities since the Nord Stream attack due to Europe's decoupling efforts from Russian gas following Russia's full-scale invasion of Ukraine. Secondly, my analysis did not find a price-increasing impact from the Nord Stream attack. The primary reason for this is that the markets already anticipated before the attack that Nord Stream would be unusable in the near future due to political reasons. Additionally, the gas import reduction due to the decoupling of Russian gas imports, which was enforced through the Nord Stream attack, was met with huge gas savings in the EU, which reduced the supply-demand gap below market anticipations. Thirdly, the coal price, included as a control variable in my gas price analysis, significantly impacted gas prices during the crisis due to simultaneous coal supply shortages. This highlights the need to consider the availability of fuel subsidies in discussions about energy supply resilience.

So, overall, the Nord Stream attack had, in the political environment in which it occurred, only a small direct impact. Its most significant consequence was that it enforced western Europe's decision to decouple from Russian gas by destroying the only direct pipeline connection between Germany and Russia, rendering pre-war import volumes impossible. However, it is beyond the scope of my paper to determine whether western Europe and Germany would have reconsidered the decision to decouple from Russian gas and return to pre-war import volumes. Further research could build upon the outlined direct impacts to answer this question. Additionally, the broader economic consequences of significant gas savings required in Europe and Germany, which may be the true costs of the challenging Nord Stream-enforced decision to decouple from Russian gas, warrant future research as long-term consequences become clearer.

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Appendix 1: Data Cleaning Gas Price Analysis

In the first appendix, I will outline the data-cleaning process of the gas price analysis in more detail than in the main text. Specifically, I will focus on the stationarity testing of the variables involved in the model and provide a comprehensive explanation regarding the exclusion of certain data points following an analysis using Cook's distance.

Given that the dataset in our gas price study comprises time-series data, I needed to assess the stationarity of each variable to confirm that data characteristics such as the mean and variance remain stable, thereby making the variables suitable for multiple linear regression analysis. To test the stationarity, I employed an Augmented Dickey-Fuller (ADF) test (Halser & Paraschiv, 2023, p.18). Table 4 displays the resulting p-values of that test. A p-value below 0.05 was deemed significant, which means that the probability of observing the data if it were non-stationary (i.e., if the null hypothesis of the presence of a unit root was true) is less than 0.05. Table 4 illustrates that only the EU-total gas inflow, the number of heating degree days, and the renewable electricity infeed were found to be stationary without modifications. All other measurements displaying no stationarity underwent differencing before ADF test 2 to achieve stationarity (Halser & Paraschiv, 2023, p.18). As it can be seen in Table 4, after the transformation all variables are stationary. Consequently, the analysis involved variables that were differenced, which is important to consider when interpreting the results.

	p-value ADF test 1	p-value ADF test 2
TTF-Gas price	0,56	0,01*
Rotterdam-Coal price	0,77	0,01*
Brent-Oil price	0,30	0,01*
gas inflow from Russia	0,44	0,01*
LNG inflow	0,17	0,01*
EU-total gas inflow	0,01*	-
gas storage filling	0,81	0,02*
heating degree days	0,01*	-
renewable electricity infeed	0,04*	-

Table 4: ADF test results

The transformed data was then tested for values that extraordinarily impact the fitting of the model. Evaluating for influential data points is crucial as their inclusion could skew the model, as outlined by Sullivan, Warkentin & Wallace, 2021, p.7). I calculated Cook's distance, which identifies outliers by comparing the difference that the inclusion of that value has on all estimations of this model and controlled the values visually (Sullivan et al., 2021, p.16). Figure 12 reveals that four observations exceed the set critical threshold of 0.1 for Cook's distance.

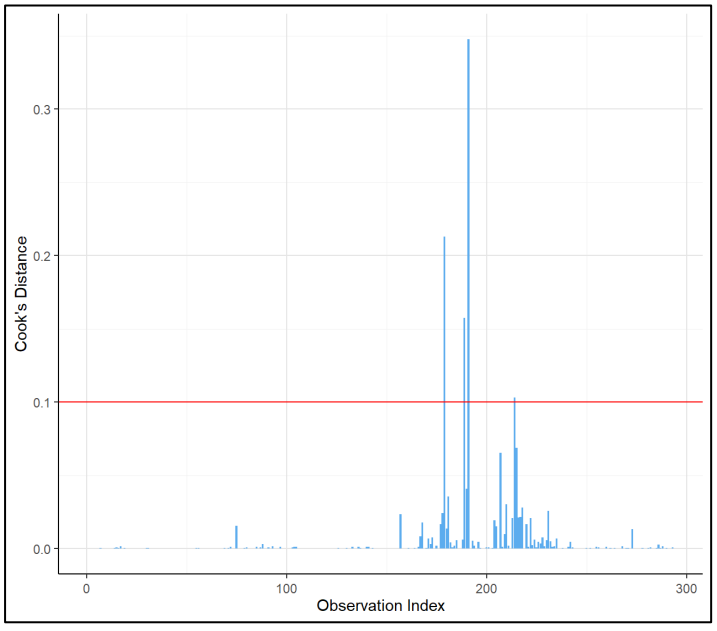


Figure 12: Cooks distance for all observations

Three of these four values are observations directly after the beginning of the Ukraine war, which I decided to exclude from the data since the gas price movements were primarily news-driven, and the independent variables saw no major movement. This exclusion is essential to prevent model overfitting to transient noise caused by the news. The other point with a Cook's distance above 0.1 is an observation after the Nord Stream attack. I decided to keep it in the data since this is our event of primary interest. The reduced dataset then includes 290 weekly observations between September 2018 and February 2024.

Appendix 2: robustness tests gas price analysis

In this appendix, I will closely examine whether the central assumptions necessary for multiple linear regression hold for my general gas price analysis model, thereby assessing the robustness of my results. The discussion will focus on four central aspects: the linearity of the correlation between the dependent and independent variables, potential multicollinearity among the independent variables, the independence of the error terms, and the normal distribution of the error term.

The assumption of linearity between the dependent and independent variables is crucial because, without it, linear regression cannot accurately estimate the relationship (Yang, 2012, p.28). To determine if this assumption holds for all variables in my model, I plotted scatterplots of each independent variable against the dependent gas price variable to visually inspect for non-linear patterns. As shown in Figure 13, the red line of best fit indicates only a small correlation for some variables, but overall, no significant non-linear patterns are observed.

Another crucial aspect that needs to be assessed is potential multicollinearity between independent variables, as it can inflate the standard errors (Taylor, 2022). To check for potential multicollinearity, I started by examining all correlations between independent variables, as shown in Table 5. Generally, the correlations are relatively low, which is positive, with the exceptions of the correlations between heating degree days and differenced gas storage, and between the industrial production index and total EU gas inflow. The negative correlation between heating degree days and differenced gas storages makes sense because extraordinary heating demand due to cold weather will be met by the gas storages.

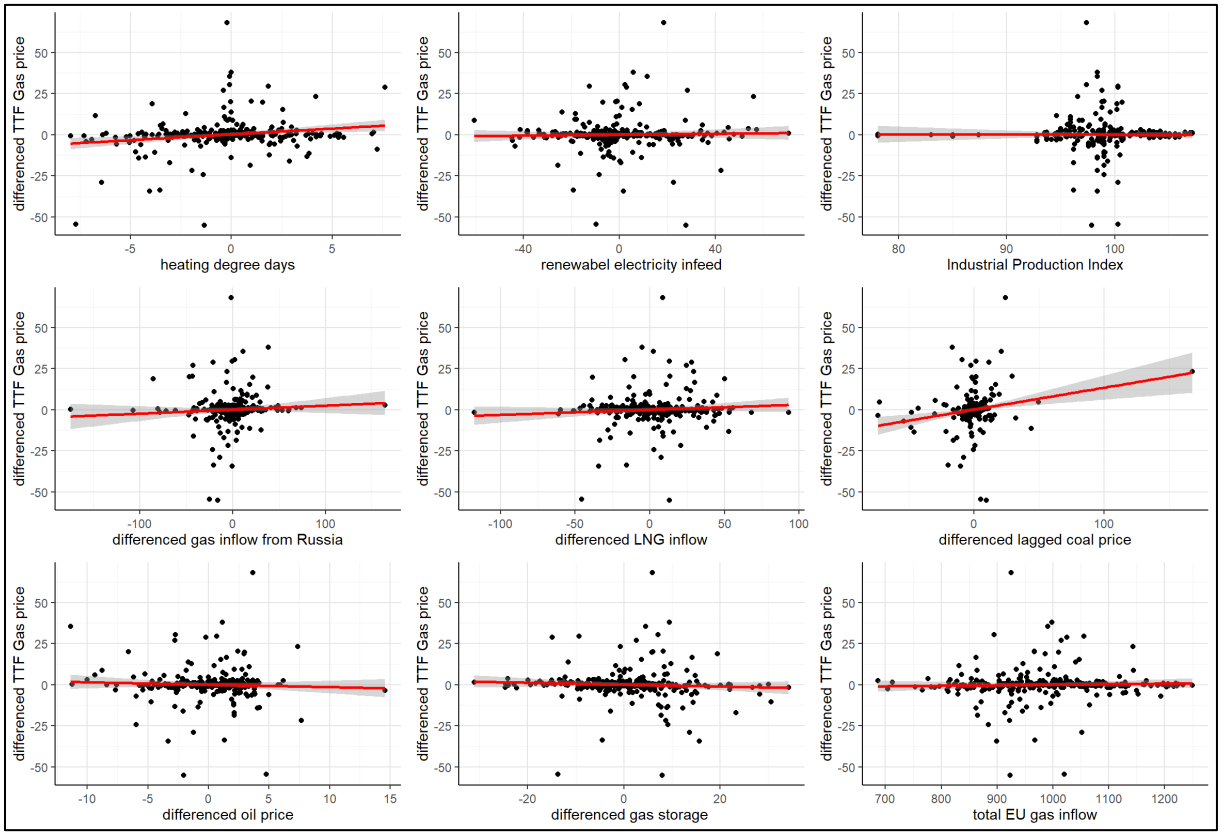


Figure 13: Scatterplots Assessing Linearity of Relationship

An increased industrial production generally increases the demand for energy and therefore the total gas imports, explaining the positive correlation between these two variables. After examining the correlations, I tested for multicollinearity by calculating the variance inflation factor for all variables. Since all values were significantly below 5, which is viewed in the literature as a threshold for potentially problematic multicollinearity (Taylor, 2022), I concluded that the assumption of no problematic multicollinearity holds for my model.

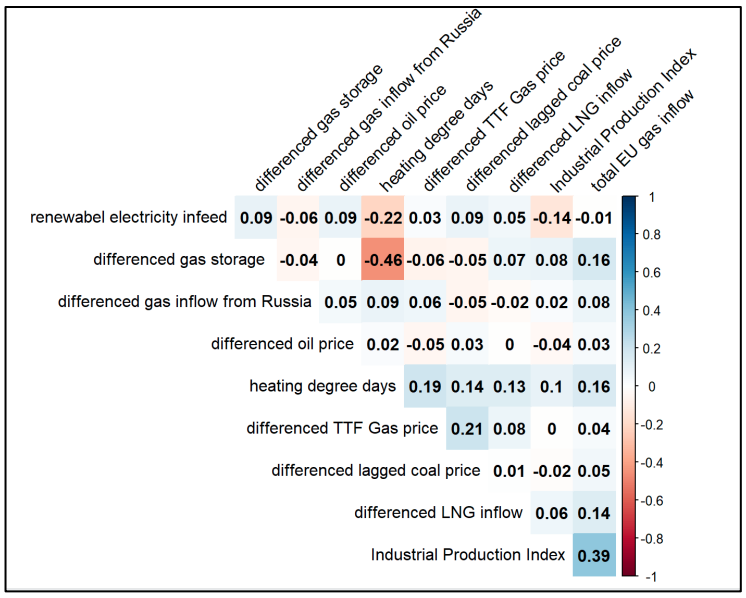


Table 5: correlation of independent variables

Regarding the error terms, they must be independent and normally distributed to satisfy the assumptions of multiple linear regression (Yang, 2012, pp. 29-33). Checking the independence of the residuals is especially important when working with time series data. I tested this assumption by calculating the autocorrelation function. As shown in Figure 14, the correlations of the residuals with each other at different time lags are consistently below the set threshold of 0.2, indicating independence. The residuals of my model are also normally distributed, as illustrated in Figure 15, so the assumption of the normal distribution of residuals is also met.

In conclusion, the critical assumptions for multiple linear regression are met in my model, indicating that the results are robust. To address the potential problem of heteroskedasticity, robust standard errors were used to calculate the t-test of my model (Eugster, 2024, p.18).

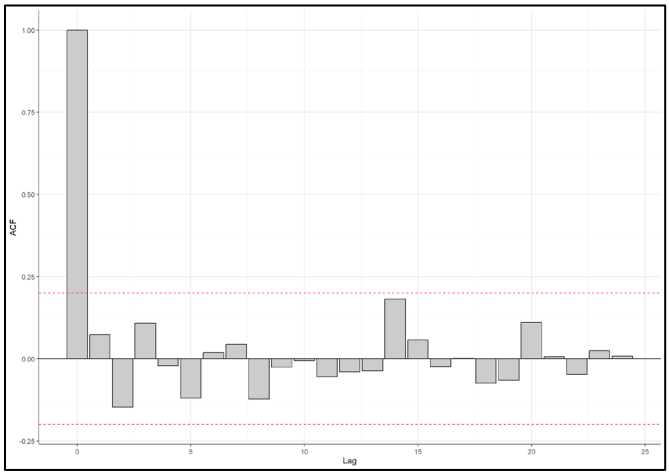


Figure 14: results of autocorrelation function

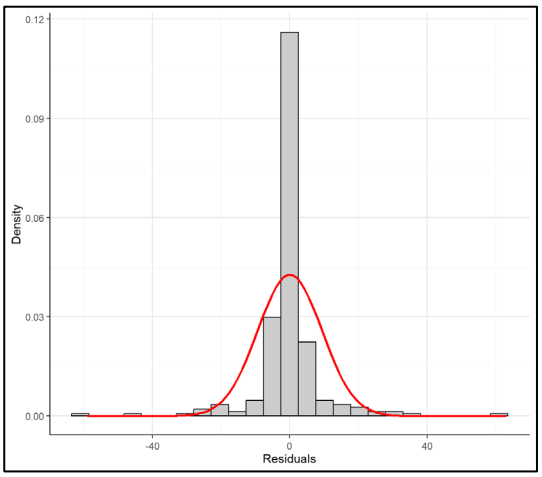


Figure 15: distribution of residuals

Appendix 3: R-Code gas-flow analysis

```
#Import Necessary Packages
library("readxl")
library(dplyr)
library(igraph)
library(tidyr)
library(ggplot2)
library(knitr)
library(kableExtra)
library(gridExtra)
library(kableExtra)
library(rmarkdown)
options(scipen = 999)

#function definition
# Function to rename the pipe index to account for two flow direction pipelines
rename_index <- function(df) {
  for (i in 2:nrow(df)) { # Start from the second row
    if (is.na(df[i, 1])) { # Check if the value is NA
      # Convert previous value to numeric, add 0.1, convert back to character
      prev_value_numeric <- as.numeric(df[i - 1, 1])
      new_prev_value <- prev_value_numeric + 0.1
      df[i - 1, 1] <- as.character(new_prev_value)
      df[i, 1] <- as.character(new_prev_value + 0.1)
    }
  }
  return(df)
}

# Function to add a from lng tag, to lng terminals
rename_lng <- function(df) {
  for (i in 1:nrow(df)) {
    if (is.na(df[i, 3])) { # Check if the value is NA
      df[i,3] <- "LNG"
    }
  }
  return(df)
}

#function to remove unused edges in max flow, and add percent of utilization
remove_unused_edges_and_store_utilization <- function(graph, utilization) {
  # Ensure that the length of the utilization vector matches the number of edges in the gra
ph
  if (length(utilization) != gsize(graph)) {
    stop("The length of the utilization vector must match the number of edges in the graph.
")
  }
  graph <- set_edge_attr(graph, "utilization", value = utilization)   # Store the utilizati
on values as an edge attribute
  edges_to_remove <- which(utilization == 0)   # Find the edges with zero utilization
  # If there are edges to remove, delete them from the graph
  if (length(edges_to_remove) > 0) {
    graph <- delete_edges(graph, E(graph)[edges_to_remove])
  }

  return(graph)
}

#function to create overview plot
plot_graph_with_utilization <- function(graph) {
  # Ensure the graph has the required attributes
  if (!"utilization" %in% names(edge.attributes(graph))) {
    stop("The graph must have an 'utilization' edge attribute.")
  }
```

```
  if (!"Number" %in% names(edge.attributes(graph))) {
    stop("The graph must have a 'Number' edge attribute.")
  }
  color_gradient <- colorRampPalette(c("blue", "red"))  # Create a color gradient from blue
(low utilization) to red (high utilization)
  # Normalize utilization values to the range [0, 1] for color mapping
  utilization_values <- E(graph)$utilization
  normalized_utilization <- (utilization_values - min(utilization_values)) /
    (max(utilization_values) - min(utilization_values))
  edge_colors <- color_gradient(100)[as.numeric(cut(normalized_utilization, breaks = 100))]
# Generate colors for each edge based on normalized utilization
  edge_labels <- paste(E(graph)$Number, round(utilization_values, 2), sep = ", ")  # Create
a combined label for each edge that includes both 'Number' and 'utilization'
  # Plot the graph with combined edge labels
  plot(graph,
       edge.label = edge_labels, # Use the combined edge labels
       edge.arrow.size = 0.5,
       vertex.color = "lightblue",
       vertex.size = 20,
       vertex.label.color = "black",
       vertex.label.cex = 0.8,
       main = "Graph Visualization",
       edge.color = edge_colors # Apply the generated colors to the edges
  )
}

#function for renaming of IEA data
generate_month_names <- function(start_date, num_columns) {
  start_month <- as.Date(start_date)  # Generiere eine Sequenz von Monaten
  month_seq <- seq(from = start_month, by = "month", length.out = num_columns)
  month_names <- format(month_seq, "%B %Y")   # Formatieren der Daten als "Monat Jahr"
  return(month_names)
}



#Graph Creation
#Load Data, clean Data, prepare data for graph creation
data <- read_excel("System Capacity Map 2024 - Capacities-Addapted.xlsx") #load ENTSOG data
data <- subset(data, select = -c(...2, ...4, ...6, `System Capacity Map 2024`)) #clean unus
ed columns
names(data) <- data[10, ] #renaming of columns
data <- data[-10, ] #deleting of name row
df_cleaned <- data[!is.na(data$`Technical physical capacity (GWh/d)`), ] #cleaning all rows
with no value for physical flow
df_graph <- subset(df_cleaned, select = c(Number, `Technical physical capacity (GWh/d)`,`Fr
om CC`,`To CC`)) #deletes all irrelevant lines for graph creation
df_graph <- rename_index(df_graph) #rename pipeline numbers, to give back flow pipelines a
separated name
df_graph <- df_graph[-184, ] #deleting of row with LNG terminal that is not yet established
df_graph <- df_graph[-120,] #deleting imaginable point in estonia 1
df_graph[136, 4] <- "DE" #add German label correctly
df_graph[140, 4] <- "DK" #add Denmark label correctly
df_graph[119, 4] <- "RUS" #add Russia label correctly
df_graph[121, 3] <- "RUS" #add Russia label correctly
df_graph <- rename_lng(df_graph) #add LNG Labels to LNG Terminals
df_graph <- df_graph %>%
  mutate(`Technical physical capacity (GWh/d)` = as.numeric(`Technical physical capacity (G
Wh/d)`)) %>% # Convert to numeric
  mutate(`Technical physical capacity (GWh/d)` = round(`Technical physical capacity (GWh/d)
`, digits = 2)) # Round to 2 decimal places
df_graph[116, 2] <- 70.5 #add missing value from ENTSOG 2021 data

#sort data in form for I graph
new_order <- names(df_graph)[c(3, 4, 2, 1)]  # Giwing the new order
```

```
df_graph <- df_graph[, new_order] #reorder the graph

#add the data not included in ENTSOG 2024 data, but necessary for the analysis based on ENT
SOG 2024 data
df_graph <- add_row(df_graph, `From CC` = "RUS", `To CC` = "DE", `Technical physical capaci
ty (GWh/d)` = 2704.40, Number =  "224")  #add Nord Stream Pipeline which is not included in
the 2024 data based on the 2021 data
df_graph <- add_row(df_graph, `From CC` = "RUS", `To CC` = "PL", `Technical physical capaci
ty (GWh/d)` = 7.3, Number =  "214")  #add Yamal Pipeline to graph data, because not include
d in 2024 because of flow stop 1
df_graph <- add_row(df_graph, `From CC` = "RUS", `To CC` = "PL", `Technical physical capaci
ty (GWh/d)` = 1027.1, Number =  "215")  #add Yamal Pipeline to graph data, because not incl
uded in 2024 because of flow stop 2
df_graph <- add_row(df_graph, `From CC` = "RUS", `To CC` = "PL", `Technical physical capaci
ty (GWh/d)` = 169.1, Number =  "216")  #add Yamal Pipeline to graph data, because not inclu
ded in 2024 because of flow stop 3
df_graph <- add_row(df_graph, `From CC` = "PL", `To CC` = "DE", `Technical physical capacit
y (GWh/d)` = 938.5, Number =  "38")  #add Yamal Pipeline to graph data, because not include
d in 2024 because of flow stop 4

#reduce data, to data relevant for our analysis
#change Belarus Labels to Russian Labels for the purpose of our analysis (all Belarus Gas c
omes from Russia, since Belarus has no relevant gas production)
df_graph <- df_graph %>%
  mutate(`From CC` = if_else(`From CC` == "BY", "RUS", `From CC`))
#change Ukrainian Labels to Russian Labels for the purpose of our analysis (all Gas from Uk
raine is Russian, since Ukraine has no relevant gas production))
df_graph <- df_graph %>%
  mutate(`From CC` = if_else(`From CC` == "UA", "RUS", `From CC`))
#filter data: only data included that have DE, AT, PL, HU, SK, UA, RO, CZ, LT, RUS
nations_vector <- c("DE", "AT", "PL", "HU", "SK", "RO", "CZ", "LT", "RUS") #reference natio
ns vector
#filtering data based on nations that the pipeline connects
df_graph_ruscon <- df_graph %>%
  filter(`From CC` %in% nations_vector & `To CC` %in% nations_vector)
#sum up all individual pipelines between two nations, to one pipline
df_graph_ruscon <- df_graph_ruscon %>%
  group_by(`From CC`, `To CC`) %>%
  summarise(`Technical physical capacity (GWh/d)` = sum(`Technical physical capacity (GWh/d
)`))
graph_ruscon <- graph_from_data_frame(df_graph_ruscon, directed = TRUE) #generate graph wit
h I graph function




#Analysis
#Max flow analysis without demand and with active Nord Stream pipelines
factor_c <- 0.03476 #define conversion factor between GWh/d to bcm per year
E(graph_ruscon)$`Technical physical capacity (GWh/d)` <- E(graph_ruscon)$`Technical physica
l capacity (GWh/d)`*factor_c #transform GWh/d in bcm per year
max_flow_nc <- max_flow(graph_ruscon, source = "RUS", target = "DE", capacity = E(graph_rus
con)$`Technical physical capacity (GWh/d)`) #calculate max flow between Russia and Germany
utilization_max <- max_flow_nc$flow/E(graph_ruscon)$`Technical physical capacity (GWh/d)`#c
alculate percent of utilization
utilization_max

##  [1] 1.0000000 0.0000000 0.0000000 0.4112511 0.0000000 0.0000000 0.0000000
##  [8] 0.0000000 0.0000000 0.0000000 1.0000000 1.0000000 0.0000000 1.0000000
## [15] 0.0000000 0.0000000 0.6934714 1.0000000 0.0000000 0.1674404 0.5496378
## [22] 0.2520309 0.5948410 0.4669866 1.0000000 0.0000000 1.0000000 0.0000000

graph_ruscon_ou <- remove_unused_edges_and_store_utilization(graph_ruscon, utilization_max)
#reduces the graph to only the utilized edges, and stores utilization values as attributes
to the edges
max_flow_nc #print max_flow value without demand and active Nord Stream pipelines
```

```
## $value
## [1] 168.1039
##
## $flow
##  [1] 27.736394  0.000000  0.000000 13.737152  0.000000  0.000000  0.000000
##  [8]  0.000000  0.000000  0.000000  1.768589  2.018513  0.000000 32.625388
## [15]  0.000000  0.000000  1.768589 94.004944  0.000000  2.018513 25.584055
## [22]  1.768589 44.727778 27.736394 13.737152  0.000000  5.022820  0.000000
##
## $cut
## + 4/28 edges from 705abf9 (vertex names):
## [1] AT ->DE PL ->DE RUS->DE SK ->CZ
##
## $partition1
## + 7/9 vertices, named, from 705abf9:
## [1] AT  HU  LT  PL  RO  RUS SK
##
## $partition2
## + 2/9 vertices, named, from 705abf9:
## [1] CZ DE
##
## $stats
## $stats$nopush
## [1] 10
##
## $stats$norelabel
## [1] 7
##
## $stats$nogap
## [1] 4
##
## $stats$nogapnodes
## [1] 1
##
## $stats$nobfs
## [1] 1

#Max flow analysis with demand baseline pre-war scenario, demand on level of 2021
#introduction of import demand overview, and graph creation
node <- c("DE", "AT", "PL", "HU", "SK", "RO", "CZ", "FR", "HOL", "BE", "SW", "IT", "SP") #a
ll relevant nations in short term
nation_names <- c("Germany", "Austria", "Poland", "Hungary", "Slovakia", "Romania", "Czechi
a", "France", "Netherlands", "Belgium", "Switzerland", "Italy", "Spain") #natino names in l
ong for graph
nations_imports <- c(55.185, 4.32, 10.545, 7.125, 3.519, 2.8, 8.7, 9.966, 10.885, 2.568, 1.
63, 29.2, 3.33 ) #nation imports 2021 in billion cubic meters
bar_graph_data <- data.frame(Nations = nation_names, Gas_imports = nations_imports) #create
data frame for import bar graph
bar_graph_data$Nations <- factor(bar_graph_data$Nations, levels = bar_graph_data$Nations[or
der(-bar_graph_data$Gas_imports)])#sort values
# create vector categories for graph
western_europe <- c("France", "Netherlands", "Belgium", "Switzerland", "Spain")
germany <- c("Germany")
transfer_nations <- c("Austria", "Poland", "Hungary", "Slovakia", "Romania", "Czechia", "It
aly")
bar_graph_data$Color <- ifelse(bar_graph_data$Nations %in% western_europe, "Western Europe"
,
                               ifelse(bar_graph_data$Nations %in% germany, "Germany", "Tran
sfer Nations"))
#create a bar graph displaying the import volumes (Figure 3)
ggplot(bar_graph_data, aes(x = Nations, y = Gas_imports, fill = Color)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = round(Gas_imports)), vjust = -0.5, color = "black", size = 3.5) +
  theme_classic() +
  labs(
       y = "Russian Gas Imports in Billion Cubic Meters",
```

```
      x = NULL) +
  scale_fill_manual(values = c("Western Europe" = "lightblue", "Germany" = "steelblue2", "T
ransfer Nations" = "salmon"),
                    labels = c("Germany", "Transfer Nations", "Western Europe")) +
  theme(axis.title.x = element_blank(),
        panel.grid.major = element_line(color = "grey90", size = 0.5),
        panel.grid.minor = element_line(color = "grey95", size = 0.25),
        legend.position = c(0.95, 0.95),
        legend.justification = c(1, 1),
        legend.box.just = "right",
        legend.margin = margin(t = -5, r = -5, b = -5, l = -5),
        legend.box.background = element_blank(),
        legend.background = element_blank())
```

```
#generate one data frame for graph creation with demand, and generate the graph
#create here direct connections from each Transfer nation to germany, this are allways util
zed first and fully since it is a direct and shorter connection to the sink, and reduce the
reby the effectiv possibel flow over the real pipline net by the demand
`From CC` <- c( "AT", "PL", "HU", "SK", "RO", "CZ", "LT") #demand on Lithuania is domestic
demand + flow to Kalinigrad based on IEA data, flow of Austria is full Austrian demand + It
alian demand, since it is the only way to Italy, because the trans Adriatic pipeline is onl
y for gas form Azerbaijan
`To CC` <- c("DE","DE", "DE", "DE", "DE", "DE", "DE" )
`Technical physical capacity (GWh/d)` <- c(33.52, 10.545, 7.125, 3.519, 2.8, 8.7,3.384) #na
tion imports 2021 in billion cubic meters
import_df_graph <- data.frame("From_CC" = `From CC`, "To_CC" = `To CC`, "Technical_physical
_capacity"= `Technical physical capacity (GWh/d)`) #combine it to one import dataframe
df_graph_ruscon <- df_graph_ruscon %>%
  rename(
    "From_CC" = `From CC`,
    "To_CC" = `To CC`,
    "Technical_physical_capacity"= `Technical physical capacity (GWh/d)`
  )
df_graph_ruscon$"Technical_physical_capacity" <- df_graph_ruscon$"Technical_physical_capaci
ty"*factor_c #transform GWh to bcm
df_graph_ruscon_demand <- bind_rows(df_graph_ruscon, import_df_graph) #combine import data
frame with cleaned data frame for graph creation
graph_ruscon_demand <- graph_from_data_frame(df_graph_ruscon_demand, directed = TRUE) #gene
rate graph with I graph function

#max flow analysis with demand and active Nord Stream pipeline (scenario 1)
max_flow_d <- max_flow(graph_ruscon_demand, source = "RUS", target = "DE", capacity = E(gra
ph_ruscon_demand)$"Technical_physical_capacity") #calculate max flow
df_results <- df_graph_ruscon_demand #store pipeline data frame in results data frame as ba
sis
max_flow_d

## $value
## [1] 227.1349
##
## $flow
##  [1] 25.8744120  0.0000000  0.0000000  5.0371520  0.0000000  0.0000000
##  [7]  0.0000000  0.0000000  0.0000000  2.6925096  1.7685888  0.0000000
## [13]  0.0000000 32.6253884  0.0000000  3.3767276  0.0000000 94.0049440
## [19] 11.5860984  3.3840000 42.8595316  0.1074904 75.1928320 59.3944120
## [25] 13.7371520  0.0000000  3.6875844  0.0000000 33.5200000 10.5450000
## [31]  7.1250000  3.5190000  2.8000000  8.7000000  3.3840000
##
## $cut
## + 9/35 edges from 70a821b (vertex names):
## [1] PL ->DE RUS->DE SK ->AT SK ->CZ PL ->DE HU ->DE SK ->DE RO ->DE LT ->DE
##
## $partition1
## + 6/9 vertices, named, from 70a821b:
```

```
## [1] HU  LT  PL  RO  RUS SK
##
## $partition2
## + 3/9 vertices, named, from 70a821b:
## [1] AT CZ DE
##
## $stats
## $stats$nopush
## [1] 16
##
## $stats$norelabel
## [1] 9
##
## $stats$nogap
## [1] 2
##
## $stats$nogapnodes
## [1] 1
##
## $stats$nobfs
## [1] 1
```

```
df_results$flow_demand <- max_flow_d$flow #generate new column with flow values in first sc
enario
df_results$flow_without_demand <- NA #create empty column to save flow values without deman
d
df_results$flow_without_demand[1:length(max_flow_nc$flow)] <- max_flow_nc$flow #save max fl
ow values with demand
value_max_flow_WE <- max_flow_d$value - sum(`Technical physical capacity (GWh/d)`) #subtrac
t the total demand from the calculated max flow, since it includes all gas that flowed trou
gh the direct connections which represent demand on the way
value_max_flow_WE
```

```
## [1] 157.5419
```

```
demand_WE <- 55.185+9.966+10.885+2.568+1.63+3.33 #sum up all demand of western Europes nati
ons
demand_WE
```

```
## [1] 83.564
```

```
utilization <- demand_WE/value_max_flow_WE #calculate the effective coverage of the demand
utilization
```

```
## [1] 0.530424
```

```
#max flow analysis with the reduction of Nord Streams capacity too 27,5 the official capaci
ty of the only intact pipeline (scenario 2)
df_graph_ruscon_demand[18,3] <- 27.5 #reduce NS capacity
graph_ruscon_demand <- graph_from_data_frame(df_graph_ruscon_demand, directed = TRUE) #gene
rate graph with I graph function
max_flow_d_aa <- max_flow(graph_ruscon_demand, source = "RUS", target = "DE", capacity = E(
graph_ruscon_demand)$"Technical_physical_capacity") #calculate max flow
max_flow_d_aa
```

```
## $value
## [1] 160.63
##
## $flow
##  [1] 25.8744120  0.0000000  0.0000000  5.0371520  0.0000000  0.0000000
##  [7]  0.0000000  0.0000000  0.0000000  2.6925096  1.7685888  0.0000000
## [13]  0.0000000 32.6253884  0.0000000  3.3767276  0.0000000 27.5000000
## [19] 11.5860984  3.3840000 42.8595316  0.1074904 75.1928320 59.3944120
## [25] 13.7371520  0.0000000  3.6875844  0.0000000 33.5200000 10.5450000
## [31]  7.1250000  3.5190000  2.8000000  8.7000000  3.3840000
##
## $cut
```

```
## + 9/35 edges from 70ace38 (vertex names):
## [1] PL ->DE RUS->DE SK ->AT SK ->CZ PL ->DE HU ->DE SK ->DE RO ->DE LT ->DE
##
## $partition1
## + 6/9 vertices, named, from 70ace38:
## [1] HU  LT  PL  RO  RUS SK
##
## $partition2
## + 3/9 vertices, named, from 70ace38:
## [1] AT CZ DE
##
## $stats
## $stats$nopush
## [1] 16
##
## $stats$norelabel
## [1] 9
##
## $stats$nogap
## [1] 2
##
## $stats$nogapnodes
## [1] 1
##
## $stats$nobfs
## [1] 1

df_results$flow_after_attac_baselinescenario <- max_flow_d_aa$flow #store results in result
s data frame
value_max_flow_WE <- max_flow_d_aa$value - sum(`Technical physical capacity (GWh/d)`) #subt
ract the total demand from the calculated max flow, since it includes all gas that flowed t
rough the direct connections which represent demand on the way
value_max_flow_WE

## [1] 91.03695

utilization <- demand_WE/value_max_flow_WE #calculate the effective coverage of the demand
utilization

## [1] 0.917913

#max flow analysis, with the full stop of transports over the yamal pipeline, which is curr
ently the case (scenario 3)
df_graph_ruscon_demand[30,3] <- 0 #reduce polish demand to 0
df_graph_ruscon_demand[21,3] <- 4.713 #reduce inflow to poland to only the Ukraine-Poland i
nflow
df_graph_ruscon_demand[14,3] <- 0 #reduce inflow from poland to germany to 0, because the p
ipline connection here is also the Yamal pipeline
graph_ruscon_demand <- graph_from_data_frame(df_graph_ruscon_demand, directed = TRUE) #gene
rate graph with I graph function
max_flow_d_aa_NY <- max_flow(graph_ruscon_demand, source = "RUS", target = "DE", capacity =
E(graph_ruscon_demand)$"Technical_physical_capacity") #calculate max flow
max_flow_d_aa_NY

## $value
## [1] 117.4596
##
## $flow
##  [1] 25.8744120  0.0000000  0.0000000  5.0371520  0.0000000  0.0000000
##  [7]  0.0000000  0.0000000  0.0000000  2.6925096  1.7685888  0.0000000
## [13]  0.0000000  0.0000000  0.0000000  0.0000000  0.0000000 27.5000000
## [19] 11.5860984  3.3840000  0.0000000  0.1074904 74.8819752 59.3944120
## [25] 13.7371520  0.0000000  0.0000000  0.0000000 33.5200000  0.0000000
## [31]  7.1250000  3.5190000  2.8000000  8.7000000  3.3840000
##
## $cut
```

```
## + 9/35 edges from 70bd753 (vertex names):
## [1] PL ->DE RUS->DE SK ->AT SK ->CZ PL ->DE HU ->DE SK ->DE RO ->DE LT ->DE
##
## $partition1
## + 6/9 vertices, named, from 70bd753:
## [1] HU  LT  PL  RO  RUS SK
##
## $partition2
## + 3/9 vertices, named, from 70bd753:
## [1] AT CZ DE
##
## $stats
## $stats$nopush
## [1] 15
##
## $stats$norelabel
## [1] 8
##
## $stats$nogap
## [1] 2
##
## $stats$nogapnodes
## [1] 1
##
## $stats$nobfs
## [1] 1

df_results$flow_after_attac_no_Yamal <- max_flow_d_aa_NY$flow #store results in results dat
a frame
value_max_flow_WE <- max_flow_d_aa_NY$value - sum(`Technical physical capacity (GWh/d)`) #s
ubtract the total demand from the calculated max flow, since it includes all gas that flowe
d trough the direct connections which represent demand on the way
value_max_flow_WE

## [1] 47.86656

utilization <- demand_WE/value_max_flow_WE #calculate the effective coverage of the demand
utilization

## [1] 1.74577

#Scenario 4 needed no max flow analysis, since only the existing Nord Stream pipe exist wit
h a capacity of 27,5 bcm, and all other connections with russia were closed
#this numbers were compared to western Europes and Germanys demand by hand



#LNG capability analysis, not explicitly mentioned in the paper, was to get a understanding
if LNG capacites were huge enough to cover demand recuction from othere sources
#filter data so that only LNG terminals to relevant eu countries are included
nations_vector <- c ("BE", "FR", "NL", "ES", "PT", "IT", "GR", "HT", "LT", "PL", "FI", "DE"
)#reference nations vector
lng_vector <- c("LNG")
df_lng <- df_graph %>%
  filter(`From CC` %in% lng_vector & `To CC` %in% nations_vector) #filtering data based on
nations that the pipline connects
total_lng_cap <- sum(df_lng$`Technical physical capacity (GWh/d)`)
total_lng_cap <- total_lng_cap*factor_c
total_lng_cap

## [1] 251.6923
```

Appendix 4: R-Code gas price analysis

```r
#Import Necessary Packages and Libarys
library(readxl)
library(dplyr)
library(igraph)
library(tidyr)
library(purrr)
library(lubridate)
library(utils)
library(httr)
library(jsonlite)
library(devtools)
library(pkgload)
library(entsog)
library(tibble)
library(readr)
library(httr)
library(tseries)
library(urca)
library(reshape2)
library(ggplot2)
library(broom)
library(stargazer)
library(corrplot)
library(lmtest)
library(nlme)
library(tseries)
library(scales)
library(sandwich)
library(forcats)
library(ggrepel)
library(car)
library(forcats)
library(gridExtra)

#Load data document
data_price_anylsis <- read.csv("data_price_analysis.csv")

#Function Definitions
# function to filter data only between certain years
data_filtering_based_on_year <- function(filter_df, start_year, end_year) {
  filter_df$Year <- as.numeric(sub("^([0-9]{4}).*", "\\1", filter_df$YearWeek))
  if (start_year != 0) {
    filter_df <- filter_df %>%
      filter(Year > start_year)  }
  if (end_year != 0) {
    filter_df <- filter_df %>%
      filter(Year < end_year)
  }
  filter_df <- select(filter_df, -Year)
  return(filter_df)
}



#AFD testing and data transformation
#AFD testing to test stationarity of the time series data
columns_to_test <- colnames(data_price_anylsis)[c(2, 3, 4,5,6,7,8,9,10,11,12,13,14,15,16)]
#specificy to test columns
adf_test_results <- list() #Initialize an empty list to store results
# Loop through each column and perform the ADF test
for(column_name in columns_to_test) {
  time_series_data <- data_price_anylsis[[column_name]]
  # Check if the column is numeric since adf.test requires numeric input
```

```
  if(is.numeric(time_series_data)) {
    adf_test_result <- adf.test(time_series_data, alternative = "stationary")
    adf_test_results[[column_name]] <- adf_test_result
  } else {
    cat(paste("Column", column_name, "is not numeric and was skipped.\n"))
  }
}

## Column NA is not numeric and was skipped.
## Column NA is not numeric and was skipped.

print(adf_test_results) # Print the results

## $Russia
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -2.3072, Lag order = 6, p-value = 0.4467
## alternative hypothesis: stationary
##
##
## $LNG
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -2.9487, Lag order = 6, p-value = 0.1764
## alternative hypothesis: stationary
##
##
## $EU.total
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -4.7623, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
##
##
## $TTF_Gas_price
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -2.0305, Lag order = 6, p-value = 0.5634
## alternative hypothesis: stationary
##
##
## $Coal_price
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -1.5272, Lag order = 6, p-value = 0.7755
## alternative hypothesis: stationary
##
##
## $Oil_price
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -2.641, Lag order = 6, p-value = 0.3061
## alternative hypothesis: stationary
##
```

```
##
## $storage_difference
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -1.4393, Lag order = 6, p-value = 0.8125
## alternative hypothesis: stationary
##
##
## $HDD
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -3.5306, Lag order = 6, p-value = 0.04015
## alternative hypothesis: stationary
##
##
## $difference_HDD
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -5.3849, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
##
##
## $renewabels_total
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -3.511, Lag order = 6, p-value = 0.04204
## alternative hypothesis: stationary
##
##
## $renewabel_difference_average
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -6.5586, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
##
##
## $Index
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -3.9532, Lag order = 6, p-value = 0.01175
## alternative hypothesis: stationary
##
##
## $CO2_price
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -0.73503, Lag order = 6, p-value = 0.9667
## alternative hypothesis: stationary

#differentiation of non stationarity data to make them stationary
data_price_anylsis$differenced_Russia <- c(NA, diff(data_price_anylsis$Russia))
data_price_anylsis$differenced_LNG <- c(NA, diff(data_price_anylsis$LNG))
```

```r
data_price_anylsis$differenced_Gas_Price <- c(NA, diff(data_price_anylsis$TTF_Gas_price))
data_price_anylsis$differenced_Coal_Price <- c(NA, diff(data_price_anylsis$Coal_price))
data_price_anylsis$differenced_Oil_Price <- c(NA, diff(data_price_anylsis$Oil_price))
data_price_anylsis$differenced_storage_difference <- c(NA, diff(data_price_anylsis$storage_
difference))
data_price_anylsis$differenced_CO2_Price <- c(NA, diff(data_price_anylsis$CO2_price))
data_price_anylsis$differenced_russia_percent <- c(NA, diff(data_price_anylsis$russia_perce
nt))
data_price_anylsis$differenced_lng_percent <- c(NA, diff(data_price_anylsis$lng_percent))
data_price_anylsis <- data_price_anylsis[-1,] #delte first row because it includes NA now

#laging of the coal variable
data_price_anylsis$lagged_coal_price <- c(NA, data_price_anylsis$differenced_Coal_Price[-le
ngth(data_price_anylsis$differenced_Coal_Price)]) #crate lagged coal variabel
data_price_anylsis <- data_price_anylsis[-1,] #delete first row

#test all transformed data again on stationarity
columns_to_test <- colnames(data_price_anylsis)[c(17,18,19,20,21,22,23,24,25,26)] # specify
columens
adf_test_results <- list()# Initialize an empty list to store results
# Loop through each column and perform the ADF test
for(column_name in columns_to_test) {
  time_series_data <- data_price_anylsis[[column_name]]
  # Check if the column is numeric since adf.test requires numeric input
  if(is.numeric(time_series_data)) {
    adf_test_result <- adf.test(time_series_data, alternative = "stationary")
    adf_test_results[[column_name]] <- adf_test_result
  } else {
    cat(paste("Column", column_name, "is not numeric and was skipped.\n"))
  }
}

## Column differenced_russia_percent is not numeric and was skipped.
## Column differenced_lng_percent is not numeric and was skipped.
## Column NA is not numeric and was skipped.
## Column NA is not numeric and was skipped.

print(adf_test_results) # Print the results

## $differenced_Gas_Price
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -6.3583, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
##
##
## $differenced_Coal_Price
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -6.0065, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
##
##
## $differenced_Oil_Price
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -5.8084, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
##
##
```

```
## $differenced_storage_difference
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -3.7285, Lag order = 6, p-value = 0.02299
## alternative hypothesis: stationary
##
##
## $differenced_CO2_Price
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -7.6694, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
##
##
## $lagged_coal_price
##
##  Augmented Dickey-Fuller Test
##
## data:  time_series_data
## Dickey-Fuller = -6.0012, Lag order = 6, p-value = 0.01
## alternative hypothesis: stationary
```

```
#create new dummy variable for the attack on Nord Stream (0 before the attack, 1 after the
attack)
cutoff_year_week = "2022-39" #set cut of point
#generate new dummy variable for begin of Ukraine war
data_price_anylsis_dummy_NSA <- data_price_anylsis %>%
  mutate(Dummy_NSA = if_else(YearWeek >= cutoff_year_week, 1, 0))
```

```
#test existence of extreme values with cooks distance
model_dummy_NSA_1 <- lm(differenced_Gas_Price ~ differenced_Russia + differenced_LNG + lagg
ed_coal_price + differenced_Oil_Price + differenced_storage_difference + difference_HDD + r
enewabels_total + Index + EU.total + Dummy_NSA, data = data_price_anylsis_dummy_NSA) #estim
ate model
cooks_d <- cooks.distance(model_dummy_NSA_1) #estimate cooks distance
cooks_df <- data.frame(Index = seq_along(cooks_d), CooksDistance = cooks_d) #store the esti
mated values in a dataframe
#plot cooks distance to inspect the values visually (Figure 12)
ggplot(cooks_df, aes(x = Index, y = CooksDistance)) +
  geom_bar(stat = "identity", fill = "steelblue2") +  # Using bars to mimic 'type="h"'
  geom_hline(yintercept = 0.1, color = "red") +  # Higher threshold
  theme_classic() +
  labs(
    x = "Observation Index",
    y = "Cook's Distance"
  ) +
  theme(
    axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 0.5),
    panel.grid.major = element_line(color = "grey90", size = 0.5),
    panel.grid.minor = element_line(color = "grey95", size = 0.25),
    legend.position = "none"
  )
```

```
#kick out rows with to high impact which were primarily the reaction to news
values_to_exclude <-c("2022-07", "2022-09", "2022-11")
data_price_anylsis_dummy_NSA_2 <- data_price_anylsis_dummy_NSA[!data_price_anylsis_dummy_NS
A$YearWeek %in% values_to_exclude, ]
```

```r
#test linearity of relationships with a scatter plot
#reduce data frame to all relevant columns
data_price_anylsis_graph <- data_price_anylsis_dummy_NSA_2[c("difference_HDD", "renewabel_difference_average", "Index", "differenced_Russia", "differenced_LNG", "differenced_Gas_Price", "lagged_coal_price", "differenced_Oil_Price", "differenced_storage_difference", "EU.total")]
#renaming to improve clarity in plot
data_price_anylsis_graph <- data_price_anylsis_graph %>% rename("renewabel electricity infeed" = "renewabel_difference_average")
data_price_anylsis_graph <- data_price_anylsis_graph %>% rename("Industrial Production Index" = "Index")
data_price_anylsis_graph <- data_price_anylsis_graph %>% rename("differenced gas inflow from Russia" = "differenced_Russia")
data_price_anylsis_graph <- data_price_anylsis_graph %>% rename("differenced LNG inflow" = "differenced_LNG")
data_price_anylsis_graph <- data_price_anylsis_graph %>% rename("differenced TTF Gas price" = "differenced_Gas_Price")
data_price_anylsis_graph <- data_price_anylsis_graph %>% rename("differenced lagged coal price" = "lagged_coal_price")
data_price_anylsis_graph <- data_price_anylsis_graph %>% rename("differenced oil price" = "differenced_Oil_Price")
data_price_anylsis_graph <- data_price_anylsis_graph %>% rename("differenced gas storage " = "differenced_storage_difference")
data_price_anylsis_graph <- data_price_anylsis_graph %>% rename("total EU gas inflow" = "EU.total")
data_price_anylsis_graph <- data_price_anylsis_graph %>% rename("heating degree days" = "difference_HDD")
# Extract column names excluding the target variable
variables <- colnames(data_price_anylsis_graph)
target_variable <- "differenced TTF Gas price"
variables <- variables[variables != target_variable]
# Create a list to store the individual plots
plot_list <- list()
# Loop through each variable to create scatter plots with lines of best fit
for (var in variables) {
  p <- ggplot(data_price_anylsis_graph, aes_string(x = paste0("`", var, "`"), y = paste0("`", target_variable, "`"))) +
    geom_point(color = "black") +
    geom_smooth(method = lm, color = "red") +
    theme_classic() +
    labs(
      x = var,
      y = target_variable,
    ) +
    theme(
      axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 0.5),
      panel.grid.major = element_line(color = "grey90", size = 0.5),
      panel.grid.minor = element_line(color = "grey95", size = 0.25),
      legend.position = "none"
    )
  plot_list[[var]] <- p
}
do.call("grid.arrange", c(plot_list, ncol = 3))  # Arrange the plots in a grid using grid.arrange (Figure 13)

#create a correlation matrix to test the data for multicollinearity
selected_columns <- c("renewabel electricity infeed", "Industrial Production Index", "differenced LNG inflow", "differenced lagged coal price", "differenced oil price", "differenced gas storage", "total EU gas inflow", "heating degree days", "differenced gas inflow from Russia") # selection of column names
correlation_matrix_selected <- cor(data_price_anylsis_graph, use = "pairwise.complete.obs")
#calculate correlation matrix
# Visualize the correlation matrix (Tabel 5)
```

```
corrplot(correlation_matrix_selected, method = "color", type = "upper", order = "hclust",
         tl.col = "black", tl.srt = 45, addCoef.col = "black", diag = FALSE)
```

```
#check multicollinearity with VIF calculation
vif_values <- vif(model_dummy_NSA_1) # Calculate VIF
print(vif_values) # Print VIF values

##             differenced_Russia                differenced_LNG
##                       1.053478                       1.059751
##              lagged_coal_price          differenced_Oil_Price
##                       1.043206                       1.018531
## differenced_storage_difference                 difference_HDD
##                       1.580448                       1.448849
##               renewabels_total                          Index
##                       1.147437                       1.297674
##                       EU.total                      Dummy_NSA
##                       2.227891                       2.037547

# Define alternative variable names for better accessibility
var_names <- c("EU.total" = "Total EU Gas Inflow",
               "differenced_Russia" = "differenced gas inflow from Russia",
               "differenced_LNG" = "differenced LNG Inflow",
               "lagged_coal_price" = "differenced lagged coal price",
               "differenced_Oil_Price" = "differenced oil price",
               "differenced_storage_difference" = "differenced gas storage",
               "difference_HDD" = "heating degree days",
               "renewabels_total" = "renewable electricity infeed",
               "Index" = "Industrial Production Index",
               "Dummy_NSA" = "Dummy North Stream attack"
)
# Renaming the column
data_price_anylsis_dummy_NSA_2 <- data_price_anylsis_dummy_NSA_2 %>%
  rename(`differenced TTF gas price` = differenced_Gas_Price)
```

```
# Define the model in different variations
model_final_1 <- lm(`differenced TTF gas price` ~ Dummy_NSA, data = data_price_anylsis_dumm
y_NSA_2)
model_final_2 <- lm(`differenced TTF gas price` ~ EU.total + differenced_Russia + differenc
ed_LNG + Dummy_NSA, data = data_price_anylsis_dummy_NSA_2)
model_final_3 <- lm(`differenced TTF gas price` ~ EU.total + differenced_Russia + differenc
ed_LNG + difference_HDD + renewabel_difference_average  + Index + Dummy_NSA, data = data_pr
ice_anylsis_dummy_NSA_2)
model_final_4 <- lm(`differenced TTF gas price` ~ EU.total + differenced_Russia + differenc
ed_LNG + lagged_coal_price + differenced_Oil_Price + differenced_storage_difference + diffe
rence_HDD + renewabel_difference_average + Index + Dummy_NSA , data = data_price_anylsis_du
mmy_NSA_2)
# Compute heteroskedasticity-robust standard errors
robust_se_1 <- vcovHC(model_final_1, type = "HC1")
robust_se_2 <- vcovHC(model_final_2, type = "HC1")
robust_se_3 <- vcovHC(model_final_3, type = "HC1")
robust_se_4 <- vcovHC(model_final_4, type = "HC1")
# Create the regression table with robust standard errors using stargazer (Tabel 3)
stargazer(model_final_1, model_final_2, model_final_3, model_final_4, type = "text",
                          title = "Regression Results",
                          align = TRUE,
                          star.cutoffs = c(0.1,0.05, 0.01, 0.001),  # Define p-value cutoff
s for stars
                          report = "vc*p",  # v: variable names, c: coefficients, p: p-valu
es
                          se = list(sqrt(diag(robust_se_1)), sqrt(diag(robust_se_2)), sqrt(
diag(robust_se_3)), sqrt(diag(robust_se_4))),  # Include robust SEs
```

```
                          add.lines = list(  # Add custom lines for additional metrics like
R-squared
                            c("R-squared",
                              format(summary(model_final_1)$r.squared, digits = 3),
                              format(summary(model_final_2)$r.squared, digits = 3),
                              format(summary(model_final_3)$r.squared, digits = 3),
                              format(summary(model_final_4)$r.squared, digits = 3))
                          ),
                          covariate.labels = var_names  # Use alternative variable names
)

##
## Regression Results
## ============================================================================================
=====================================
##                                                                          Dependent variabl
e:
##                                    ----------------------------------------------------
---------------------------------------
##                                                                           `differenced TTF gas
price`
##                                              (1)                   (2)
(3)                     (4)
## --------------------------------------------------------------------------------------------
---------------------------------------
## Total EU Gas Inflow                                          -0.011                 -
0.010                  -0.013
##                                                             p = 0.123             p =
0.143               p = 0.185
##
## differenced gas inflow from Russia                           0.031*                 0
.026                  0.033*
##                                                             p = 0.099             p =
0.158               p = 0.094
##
## differenced LNG Inflow                                         0.037                 0
.027                  0.026
##                                                             p = 0.122             p =
0.207               p = 0.198
##
## differenced lagged coal price
0.111***
##
p = 0.004
##
## differenced oil price
-0.206
##
p = 0.367
##
## differenced gas storage
0.084
##
p = 0.407
##
## heating degree days                                                                    0.
651**                 0.711*
##                                                                                       p =
0.020               p = 0.074
##
## renewable electricity infeed                                                           0
.024                  0.017
##                                                                                       p =
0.387               p = 0.502
##
## Industrial Production Index                                                            -
```

```
0.074                  -0.078
##                                                                                  p =
0.116               p = 0.102
##
## Dummy North Stream attack                -3.003***             -4.783**               -4
.168**                 -4.507*
##                                          p = 0.009             p = 0.016             p =
0.031               p = 0.055
##
## Constant                                   0.823                12.250                18
.824**                22.372*
##                                          p = 0.245             p = 0.118             p =
0.036               p = 0.088
##
## ------------------------------------------------------------------------------------------
--------------------------------------
## R-squared                                 0.0179                0.0367                 0
.0657                  0.104
## Observations                               290                   290
290                     290
## R2                                        0.018                 0.037                 0
.066                   0.104
## Adjusted R2                               0.014                 0.023                 0
.043                   0.072
## Residual Std. Error                  9.774 (df = 288)      9.731 (df = 285)       9.634
(df = 282)        9.485 (df = 279)
## F Statistic                         5.247** (df = 1; 288) 2.715** (df = 4; 285) 2.833***
(df = 7; 282) 3.241**** (df = 10; 279)
## ==========================================================================================
======================================
## Note:
*p<0.1; **p<0.05; ***p<0.01
```

```r
#test Normal distribution for Robustness test
residuals <- resid(model_final_4) # Calculate residuals
residuals_data <- data.frame(Residuals = residuals) # Create a data frame with the residual
mean_residuals <- mean(residuals) # Calculate mean
sd_residuals <- sd(residuals) # calculate standard deviation of the residuals
# Plot histogram with density curve using of residuals to test normal distribution visually
(Figure 15)
ggplot(residuals_data, aes(x = Residuals)) +
  geom_histogram(aes(y = ..density..), bins = nclass.scott(residuals), fill = "grey80", col
or = "black") +
  stat_function(fun = dnorm, args = list(mean = mean_residuals, sd = sd_residuals), color =
"red", size = 1) +
  theme_classic() +
  labs(
    x = "Residuals",
    y = "Density",
  ) +
  theme(
    axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 0.5),
    panel.grid.major = element_line(color = "grey90", size = 0.5),
    panel.grid.minor = element_line(color = "grey95", size = 0.25),
    legend.position = "none"  # No legend needed for this plot
  )
```

```r
#check independence of residuals
acf_values <- acf(residuals, plot = FALSE) # Calculate ACF values
acf_data <- data.frame(Lag = acf_values$lag, ACF = acf_values$acf) # Create a dataframe for
ACF values
# Plot ACF (Figure 14)
ggplot(acf_data, aes(x = Lag, y = ACF)) +
  geom_bar(stat = "identity", position = "dodge", fill = "grey80", color = "black") +
```

```
  geom_hline(yintercept = 0, color = "black") +
  geom_hline(yintercept = 0.2, linetype = "dashed", color = "red") +
  geom_hline(yintercept = -0.2, linetype = "dashed", color = "red") +
  theme_classic() +
  labs(
    x = "Lag",
    y = "ACF"
  ) +
  theme(
    axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 0.5),
    panel.grid.major = element_line(color = "grey90", size = 0.5),
    panel.grid.minor = element_line(color = "grey95", size = 0.25),
    legend.position = "none"  # No legend needed for this plot
  )

#create a graph displaying the coal, gas and oil price
#create a time index for the graphs
# Extract the year and week into separate columns
data_price_anylsis$Index_Time <- as.numeric(as.factor(data_price_anylsis$YearWeek))
data_price_anylsis <- data_price_anylsis %>%
  mutate(Year = as.numeric(substr(YearWeek, 1, 4)),  # Extract year
         Week = as.numeric(substr(YearWeek, 6, 7)))  # Extract week
# Filter for rows where the 'Week' is 1 to get the start of each year
start_of_years <- data_price_anylsis %>%
  filter(Week == 1) %>%
  .$Year %>%
  unique()
# Create breaks for the start of each year
breaks_start_of_years <- data_price_anylsis %>%
  filter(Year %in% start_of_years & Week == 1) %>%
  pull(Index_Time)
labels_start_of_years <- as.character(start_of_years) # Create labels for the breaks as jus
t the year
#plot the coal, gas and oil price (Figure 6)
ggplot(data_price_anylsis, aes(x = Index_Time)) +
  geom_line(aes(y = TTF_Gas_price, colour = "TTF Gas Price"), size = 0.65) +
  geom_line(aes(y = Coal_price, colour = "Coal Price"), size = 0.65) +
  geom_line(aes(y = Oil_price, colour = "Oil Price"), size = 0.65) +
  theme_classic() +
  labs(
    x = "Year",
    y = "weekly average price in USD") +
  scale_color_manual(
    values = c("TTF Gas Price" = "steelblue2", "Coal Price" = "salmon", "Oil Price" = "blac
k"),
    labels = c("Coal Price", "Oil Price", "TTF Gas Price"),
    guide = guide_legend(title = NULL)
  ) +
  scale_x_continuous(
    breaks = breaks_start_of_years,  # Use breaks for the start of each year
    labels = labels_start_of_years   # Use year as labels
  ) +
  theme(
    axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 0.5),  # Set angle to 0 for
horizontal labels
    panel.grid.major = element_line(color = "grey90", size = 0.5),
    panel.grid.minor = element_line(color = "grey95", size = 0.25),
    legend.position = c(0.95, 0.95),
    legend.justification = c(1, 1),
    legend.box.just = "right",
    legend.margin = margin(t = -5, r = -5, b = -5, l = -5),
    legend.box.background = element_blank(),
    legend.background = element_blank()
  )
```

```
#create a graph that shows the EU, LNG and Russian inflows as shares
data_price_anylsis$Russia_cumulative = data_price_anylsis$Russia # Calculate the cumulative
shares for Russia
data_price_anylsis$LNG_cumulative = data_price_anylsis$Russia + data_price_anylsis$LNG # Ca
lculate the cumulative shares for LNG
# plot the shares of EU, LNG and Russian gas inflows (Figure 7)
ggplot(data_price_anylsis, aes(x = Index_Time)) +
  geom_ribbon(aes(ymin = 0, ymax = EU.total, fill = "EU Total"), alpha = 0.2) +
  geom_ribbon(aes(ymin = 0, ymax = Russia_cumulative, fill = "Russia"), alpha = 0.4) +
  geom_ribbon(aes(ymin = 0, ymax = LNG_cumulative, fill = "LNG"), alpha = 0.4) +
  geom_line(aes(y = EU.total, colour = "EU Total"), size = 0.65) +
  geom_line(aes(y = Russia_cumulative, colour = "Russia"), size = 0.65) +
  geom_line(aes(y = LNG_cumulative, colour = "LNG"), size = 0.65) +
  theme_classic() +
  labs(
    y = "Weekly Gas inflow in EU in million cubic meters",
    x = "Year"
  ) +
  scale_color_manual(
    values = c("EU Total" = "blue", "Russia" = "salmon", "LNG" = "steelblue2"),
    labels = c("EU Total","LNG Imports", "Imports from Russia"),
    guide = guide_legend(title = NULL)
  ) +
  scale_fill_manual(
    values = c("EU Total" = "lightblue", "Russia" = "lightsalmon", "LNG" = "lightsteelblue"
),
    labels = c("EU Total", "Russia", "LNG"),
    guide = FALSE
  ) +
  scale_x_continuous(
    breaks = breaks_start_of_years,
    labels = labels_start_of_years
  ) +
  theme(
    axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 0.5),
    panel.grid.major = element_line(color = "grey90", size = 0.5),
    panel.grid.minor = element_line(color = "grey95", size = 0.25),
    legend.position = c(0.95, 0.95),
    legend.justification = c(1, 1),
    legend.box.just = "right",
    legend.margin = margin(t = -5, r = -5, b = -5, l = -5),
    legend.box.background = element_blank(),
    legend.background = element_blank()
  )

#visualization of the development of the storage volume
ggplot(data_price_anylsis, aes(x = Index_Time)) +
  geom_area(aes(y = storage_difference), fill = "lightblue", alpha = 0.5) +
  geom_line(aes(y = storage_difference, colour = "Storage Difference"), size = 0.65) +
  geom_hline(yintercept = 0, color = "salmon", size = 1.2) +
  theme_classic() +
  labs(
    y = "Difference from 3 year gasstorage average in million cubic meters",
    x = "Year"
  ) +
  scale_color_manual(
    values = c("Storage Difference" = "steelblue2"),
    guide = guide_legend(title = NULL)
  ) +
  scale_x_continuous(
    breaks = breaks_start_of_years,
    labels = labels_start_of_years
  ) +
  theme(
    axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 0.5),
    panel.grid.major = element_line(color = "grey90", size = 0.5),
```

```
    panel.grid.minor = element_line(color = "grey95", size = 0.25),
    legend.position = c(0.95, 0.95),
    legend.justification = c(1, 1),
    legend.box.just = "right",
    legend.margin = margin(t = -5, r = -5, b = -5, l = -5),
    legend.box.background = element_blank(),
    legend.background = element_blank()
  )

# visualizing the gas storage volume with the gas price (Figure 9)
# Calculate the range of TTF_Gas_price and storage_difference to adjust the scale properly
max_storage <- max(data_price_anylsis$storage_difference, na.rm = TRUE)
min_storage <- min(data_price_anylsis$storage_difference, na.rm = TRUE)
max_price <- max(data_price_anylsis$TTF_Gas_price, na.rm = TRUE)
min_price <- min(data_price_anylsis$TTF_Gas_price, na.rm = TRUE)
scale_factor <- (max_storage - min_storage) / (max_price - min_price) # Create scaling fact
or to align gas prices on the secondary axis
#plot gas storage and gas price
ggplot(data_price_anylsis, aes(x = Index_Time)) +
  geom_area(aes(y = storage_difference), fill = "lightblue", alpha = 0.5) +
  geom_line(aes(y = storage_difference, colour = "storage difference 3 year average"), size
= 0.65) +
  geom_hline(yintercept = 0, color = "salmon", size = 1.2) +
  geom_line(aes(y = TTF_Gas_price * scale_factor, colour = "weekly average TTF Gas Price"),
size = 0.5) +
  scale_y_continuous(
    "Difference from 3 year gas storage average in million cubic meters",
    sec.axis = sec_axis(~ . / scale_factor, name = "TTF Gas Price ($)")) +
  theme_classic() +
  labs(x = "Year") +
  scale_color_manual(
    values = c("storage difference 3 year average" = "steelblue2",
               "weekly average TTF Gas Price" = "black"),
    guide = guide_legend(title = NULL)
  ) +
  scale_x_continuous(
    breaks = breaks_start_of_years,
    labels = labels_start_of_years
  ) +
  theme(
    axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 0.5),
    panel.grid.major = element_line(color = "grey90", size = 0.5),
    panel.grid.minor = element_line(color = "grey95", size = 0.25),
    legend.position = c(0.35, 0.95),
    legend.justification = c(1, 1),
    legend.box.just = "right",
    legend.margin = margin(t = -5, r = -5, b = -5, l = -5),
    legend.box.background = element_blank(),
    legend.background = element_blank()
  )

#create pipline capacity and utilization compare graph
lng_import_data <- read.csv("LNG_Import_data.csv") #import Bruegel data set
lng_import_data$dates <- dmy(lng_import_data$dates)
# Define time frames for pipeline capacities
time_frame1_start <- as.Date("2018-08-01")
time_frame1_end <- as.Date("2022-05-01")
time_frame2_start <- as.Date("2022-05-01")
time_frame2_end <- as.Date("2022-09-01")
time_frame3_start <- as.Date("2022-09-01")
time_frame3_end <- as.Date("2024-01-01")
min_date <- as.Date("2022-01-01")
# Add a new column with capacities based on the time frames
lng_import_data <- lng_import_data %>%
  mutate(Max_Flow = case_when(
    dates >= time_frame1_start & dates <= time_frame1_end ~ 18917,
```

```
    dates >= time_frame2_start & dates <= time_frame2_end ~ 15325,
    dates >= time_frame3_start & dates <= time_frame3_end ~ 9788,
    TRUE ~ 0 # default case if date doesn't fall in any time frame
  ))
lng_import_data <- lng_import_data[c("dates", "Russia", "Max_Flow")] #reduce data frame onl
y to relevant columns
#group data by months
lng_import_monthly_data <- lng_import_data %>%
  mutate(month = floor_date(dates, "month")) %>%
  group_by(month) %>%
  summarise(
    across(-c(dates, Max_Flow), sum, na.rm = TRUE),
    max_flow_median = median(Max_Flow, na.rm = TRUE)
  )
# Filter the data for dates after 2021-12-01
filtered_data <- lng_import_monthly_data %>%
  filter(month > as.Date("2021-12-01")) %>%
  filter(month < as.Date("2023-12-01"))
#plot pipeline capacity and utilization compare graph (Figure 4)
ggplot(filtered_data, aes(x = month)) +
  geom_line(aes(y = Russia, colour = "EU Gas Imports from Russia"), size = 1.25) +
  geom_line(aes(y = max_flow_median, colour = "Max Flow"), size = 1.25) +
  geom_vline(xintercept = as.Date("2022-05-01"), linetype = "dashed", color = "black", size
= 1) +
  geom_vline(xintercept = as.Date("2022-09-01"), linetype = "dashed", color = "black", size
= 1) +
  annotate("text", x = as.Date("2022-05-01"), y = Inf, label = "Yamal Closing", hjust = -0.
1, vjust = 2) +
  annotate("text", x = as.Date("2022-09-01"), y = Inf, label = "Nord Stream Attack", hjust
= -0.1, vjust = 2) +
  labs(x = "Date", y = "Value in Million Cubic Meters", colour = "Legend") +
  theme_classic() +
  scale_color_manual(values = c("EU Gas Imports from Russia" = "lightblue", "Max Flow" = "s
almon")) +
  scale_x_date(limits = c(min_date, NA), expand = c(0, 0)) + # Turn off expansion for x-axi
s
  theme(
    axis.title.x = element_blank(),
    panel.grid.major = element_line(color = "grey90", size = 0.5),
    panel.grid.minor = element_line(color = "grey95", size = 0.25),
    legend.position = c(0.9, 0.9),
    legend.justification = c("right", "top"),
    legend.box.just = "right",
    legend.box.margin = margin(t = -20, r = -20)
  )

#Segmented Regression analysis visualization
#create data to integrate segmented regression into data
Segmented_Regression_Data <- data_price_anylsis_dummy_NSA_2 #copy data
Segmented_Regression_Data$Time <- seq_along(Segmented_Regression_Data$YearWeek) #generate T
ime Variable
Segmented_Regression_Data$post_intervention_time = Segmented_Regression_Data$Time * data_pr
ice_anylsis_dummy_NSA_2$Dummy_NSA #create post intervention time
# Extract the year and week into separate columns
Segmented_Regression_Data$Index_Time <- as.numeric(as.factor(Segmented_Regression_Data$Year
Week))
Segmented_Regression_Data <- Segmented_Regression_Data %>%
  mutate(Year = as.numeric(substr(YearWeek, 1, 4)),  # Extract year
         Week = as.numeric(substr(YearWeek, 6, 7)))  # Extract week
# Filter for rows where the 'Week' is 1 to get the start of each year
start_of_years <- Segmented_Regression_Data %>%
  filter(Week == 1) %>%
  .$Year %>%
  unique()
# Create breaks for the start of each year
breaks_start_of_years <- Segmented_Regression_Data %>%
```

```
  filter(Year %in% start_of_years & Week == 1) %>%
  pull(Index_Time)
labels_start_of_years <- as.character(start_of_years) # Create labels for the breaks as jus
t the year
model_segmented_Regression <- lm(TTF_Gas_price  ~ Index_Time + Dummy_NSA + post_interventio
n_time, data = Segmented_Regression_Data) #create segmented regression model
summary(model_segmented_Regression) #summarize results

##
## Call:
## lm(formula = TTF_Gas_price ~ Index_Time + Dummy_NSA + post_intervention_time,
##     data = Segmented_Regression_Data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -43.178 -28.490  -3.245  15.633 196.640
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)            -20.92302    4.82133  -4.340 1.98e-05 ***
## Index_Time               0.58765    0.03871  15.183  < 2e-16 ***
## Dummy_NSA              403.19417   47.94990   8.409 1.99e-15 ***
## post_intervention_time  -1.87783    0.19182  -9.789  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 35.22 on 286 degrees of freedom
## Multiple R-squared:  0.4997, Adjusted R-squared:  0.4944
## F-statistic:  95.2 on 3 and 286 DF,  p-value: < 2.2e-16

#create a graph to visualize the results
Segmented_Regression_Data$fitted_values <- predict(model_segmented_Regression)# Create a ne
w column for the fitted values from the model
label_data <- Segmented_Regression_Data[Segmented_Regression_Data$Index_Time %in% c(180, 19
2, 216), ]
label_data$custom_label <- c("Russian suppy shortcut", "Beginn of Ukraine war", "Delivery s
top trough North Stream")  # Customize these texts as needed
# Plot the segemented regression results (Figure 10)
ggplot(Segmented_Regression_Data, aes(x = Index_Time, y = TTF_Gas_price)) +
  geom_line(aes(color = as.factor(Dummy_NSA)), alpha = 0.6) + # Data points, colored by int
ervention status
  geom_line(aes(y = fitted_values), color = "salmon") + # Fitted line
  labs(
    x = "Year", y = "TTF Gas Price in USD") +
  scale_color_manual(values = c("blue", "steelblue"), labels = c("Pre North-Stream attack",
"Post North-Stream attack")) +
  theme_classic() +
  scale_x_continuous(
    breaks = breaks_start_of_years,  # Use breaks for the start of each year
    labels = labels_start_of_years   # Use year as labels
  ) +
  theme(
    axis.text.x = element_text(angle = 0, vjust = 0.5, hjust = 0.5),
    panel.grid.major = element_line(color = "grey90", size = 0.5),
    panel.grid.minor = element_line(color = "grey95", size = 0.25),
    legend.position = c(0.95, 0.95),
    legend.justification = c(1, 1),
    legend.box.just = "right",
    legend.margin = margin(t = -5, r = -5, b = -5, l = -5),
    legend.box.background = element_blank(),
    legend.background = element_blank(),
    legend.title = element_blank() # Removes legend headline
  )
```

Declaration of Authorship

I hereby declare,

- that I have written this thesis independently,
- that I have written the thesis using only the aids specified in the index;
- that all parts of the thesis produced with the help of aids have been precisely declared;
- that I have mentioned all sources used and cited them correctly according to established academic citation rules;
- that I have acquired all immaterial rights to any materials I may have used, such as images or graphics, or that these materials were created by me;
- that the topic, the thesis or parts of it have not already been the object of any work or examination of another course, unless this has been expressly agreed with the faculty member in advance and is stated as such in the thesis;
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By submitting this thesis, I confirm through my conclusive action that I am submitting the Declaration of Authorship, that I have read and understood it, and that it is true.

Declaration of AI-Tools

I utilized GPT-4 and GPT-3.5 by OpenAI, also known as ChatGPT, for writing this thesis. GPT-4 and GPT-3.5 are chatbots that can generate text in response to your prompts and fulfil various tasks. My primary utilization of these tools was for assistance in coding my R analysis and for correcting spelling and grammar. For assistance with my coding process, I used GPT-4 and GPT-3.5 to find functions and packages, code clearly defined parts, and assist in debugging my code. Example prompts are:

«Name me a function that calculates the max flow in a graph between two nods, whit respect to edge capacity.»
«Write me a code that displays the developments oft two columns of a data frame in a graph, with a red reference line for a value of 0.»
«Explain me what the variable min_data does in the following code, and why it is not defined. Code:»

For proofreading, I utilized GPT-4 and GPT-3.5 to provide feedback on grammar and spelling. I ensured that the text was not changed and that the content I wrote stayed the same. It was therefore used as an enhanced spelling correction program. This is an example prompt that I used:

«Take the text, and correct spelling and grammar. Keep the text in the exact original form, and give corrections if you think something should be changed in brackets behind the old sentence and only there. Do this for the whole text with equal attention. Most important is that there should be NO changes in the text, only recommended changes in the brackets. Here is the text:»

For the purpose of research, I used Perplexity AI to a small extent with limited success. Perplexity AI tries to answer questions as well as possible and provides all sources for its answers. I then always utilized and cited the primary sources. It was therefore used as an advanced search engine. This is an example prompt:

«Explain me the concept of Cooks-Distance»
